

THE SLEEP RESET

A Woman's Guide to Understanding
Insomnia, Breaking the Cycle, and
Getting the Rest You Deserve

Hannah Wells

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A Woman's Guide to Understanding Insomnia, Breaking the Cycle, and
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For every woman who has been told it's all in her head
—
your body was telling the truth.
This book is for learning to listen,
and for finding your way back to yourself.

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Foreword

FOREWORD: WHY SLEEP MATTERS MORE THAN YOU THINK
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SLEEP IS THE MOST OVERLOOKED PILLAR OF HUMAN HEALTH. WE
SPEND ROUGHLY A THIRD OF OUR LIVES DOING IT, YET MOST OF US
UNDERSTAND LESS ABOUT IT THAN WE DO ABOUT NUTRITION OR
EXERCISE. WHEN SLEEP WORKS, WE BARELY NOTICE IT. WHEN IT
BREAKS, IT BREAKS EVERYTHING. I WROTE THIS BOOK BECAUSE
SLEEP DISORDERS ARE ONE OF THE MOST COMMON YET MOST
MISUNDERSTOOD HEALTH CHALLENGES OF OUR TIME. THEY AFFECT
PEOPLE OF EVERY AGE, BACKGROUND, AND LIFESTYLE. THEY
MASQUERADE AS OTHER CONDITIONS — DEPRESSION, ANXIETY,
CHRONIC FATIGUE, BRAIN FOG, EVEN RELATIONSHIP PROBLEMS.
AND FOR TOO LONG, THEY HAVE BEEN DISMISSED AS TRIVIAL. "JUST
GO TO BED EARLIER." "STOP DRINKING COFFEE." "HAVE YOU TRIED
CHAMOMILE TEA?" IF YOU HAVE HEARD THESE WORDS WHILE LYING
AWAKE AT 3 A.M., YOU KNOW HOW HOLLOW THEY SOUND. SLEEP
DISORDERS ARE NOT A CHOICE. THEY ARE NOT A SIGN OF WEAKNESS.
THEY ARE COMPLEX MEDICAL CONDITIONS THAT DESERVE THE SAME
SERIOUSNESS AS ANY OTHER HEALTH ISSUE. WHO THIS BOOK IS
FOR THIS BOOK IS FOR ANYONE WHO STRUGGLES WITH SLEEP. IT IS
FOR THE PERSON WHO FALLS ASLEEP EASILY BUT WAKES UP FOUR
TIMES A NIGHT. IT IS FOR THE PERSON WHO LIES IN BED FOR HOURS,
MIND RACING, WHILE THE CLOCK TICKS TOWARD DAWN. IT IS FOR
THE PARTNER OF SOMEONE WHO SNORES SO LOUDLY THAT SEPARATE
BEDROOMS HAVE BECOME A MARITAL SURVIVAL STRATEGY. IT IS FOR
THE PARENT OF A CHILD WHO SLEEPWALKS, THE SHIFT WORKER

WHOSE BODY CLOCK NEVER ADJUSTS, AND THE SENIOR WHO HAS NOT HAD A DEEP SLEEP IN YEARS. IF YOU HAVE EVER FELT THAT SLEEP IS A BATTLE YOU ARE LOSING, THIS BOOK IS FOR YOU. WHAT THIS BOOK IS NOT THIS BOOK IS NOT A COLLECTION OF QUICK FIXES. IT DOES NOT PROMISE THAT YOU WILL SLEEP PERFECTLY AFTER READING ONE CHAPTER. IT IS NOT A SUBSTITUTE FOR MEDICAL DIAGNOSIS OR TREATMENT. SLEEP DISORDERS ARE MEDICAL CONDITIONS, AND SOME OF THEM — PARTICULARLY SLEEP APNEA — REQUIRE PROFESSIONAL INTERVENTION. WHAT THIS BOOK OFFERS IS UNDERSTANDING. IT EXPLAINS THE SCIENCE OF SLEEP IN CLEAR, ACCESSIBLE LANGUAGE. IT WALKS YOU THROUGH THE MAJOR SLEEP DISORDERS, THEIR SYMPTOMS, CAUSES, AND TREATMENTS. IT PROVIDES PRACTICAL STRATEGIES THAT ARE GROUNDED IN RESEARCH AND TESTED IN REAL LIFE. AND IT GIVES YOU A ROADMAP FOR BUILDING A PERSONALIZED SLEEP RECOVERY PLAN. A NOTE ON MEDICAL ADVICE I AM NOT A DOCTOR. THE INFORMATION IN THIS BOOK IS BASED ON PUBLISHED RESEARCH, CLINICAL GUIDELINES, AND THE EXPERIENCES OF COUNTLESS PEOPLE WHO HAVE NAVIGATED SLEEP DISORDERS. IT IS INTENDED TO EDUCATE AND EMPOWER, NOT TO DIAGNOSE OR TREAT. IF YOU SUSPECT YOU HAVE A SLEEP DISORDER, PLEASE CONSULT A HEALTHCARE PROFESSIONAL. SLEEP MEDICINE IS A SPECIALIZED FIELD, AND THE RIGHT DIAGNOSIS CAN BE LIFE-CHANGING. HOW TO USE THIS BOOK THE BOOK IS ORGANIZED INTO THREE PARTS. THE FIRST PART (CHAPTERS 1-2) EXPLAINS WHAT SLEEP IS AND WHY IT MATTERS. THE SECOND PART (CHAPTERS 3-8) COVERS THE MAJOR SLEEP DISORDERS IN DETAIL. THE THIRD PART (CHAPTERS 9-11) FOCUSES ON TREATMENT AND RECOVERY. EACH CHAPTER ENDS WITH REFLECTION QUESTIONS OR PRACTICAL EXERCISES. THESE ARE INVITATIONS TO APPLY WHAT

YOU HAVE LEARNED, NOT HOMEWORK ASSIGNMENTS. TAKE WHAT RESONATES AND LEAVE THE REST. A NOTE ON LANGUAGE THROUGHOUT THIS BOOK, I USE THE TERM "SLEEP DISORDER" BROADLY TO INCLUDE CONDITIONS LIKE INSOMNIA, SLEEP APNEA, RESTLESS LEGS SYNDROME, CIRCADIAN RHYTHM DISORDERS, AND PARASOMNIAS. THESE ARE DISTINCT CONDITIONS WITH DIFFERENT CAUSES AND TREATMENTS, BUT THEY SHARE ONE THING IN COMMON: THEY ROB PEOPLE OF RESTORATIVE SLEEP. I ALSO USE THE FIRST PERSON OCCASIONALLY, DRAWING FROM MY OWN JOURNEY WITH SLEEP. THIS IS NOT BECAUSE MY EXPERIENCE IS UNIVERSAL — IT IS NOT — BUT BECAUSE I BELIEVE THAT VULNERABILITY CREATES CONNECTION, AND CONNECTION IS PART OF HEALING. BEFORE WE BEGIN IF YOU ARE READING THIS BOOK BECAUSE YOU ARE EXHAUSTED — TRULY, BONE-DEEP EXHAUSTED — I WANT YOU TO KNOW THAT THERE IS HOPE. SLEEP DISORDERS ARE TREATABLE. PEOPLE RECOVER. YOUR RELATIONSHIP WITH SLEEP CAN BE TRANSFORMED. THE FIRST STEP IS UNDERSTANDING. LET US BEGIN THERE. THE SLEEP-DEPRIVED SOCIETY WE LIVE IN A CULTURE THAT HAS SYSTEMATICALLY DEVALUED SLEEP. THE NINETEENTH CENTURY GAVE US THE LIGHT BULB, AND THE TWENTIETH GAVE US THE TWENTY-FOUR-HOUR NEWS CYCLE, THE INTERNET, AND THE EXPECTATION OF CONSTANT AVAILABILITY. SLEEP BECAME THE CASUALTY OF PROGRESS. WE WEAR OUR EXHAUSTION AS A BADGE OF HONOR, EQUATING SLEEP DEPRIVATION WITH DEDICATION AND PRODUCTIVITY. THE IRONY IS THAT SLEEP DEPRIVATION MAKES US LESS PRODUCTIVE, NOT MORE. A WELL-RESTED BRAIN WORKS FASTER, THINKS MORE CLEARLY, MAKES BETTER DECISIONS, AND REGULATES EMOTIONS MORE EFFECTIVELY. THE TIME WE STEAL FROM SLEEP IS NOT TIME GAINED. IT IS TIME

BORROWED AT A HIGH INTEREST RATE. THIS CULTURAL NORMALIZATION OF SLEEP DEPRIVATION IS ONE OF THE BIGGEST BARRIERS TO TREATING SLEEP DISORDERS. WHEN EVERYONE AROUND YOU IS TIRED TOO, IT IS EASY TO BELIEVE THAT EXHAUSTION IS NORMAL. IT IS NOT. THE EPIDEMIC OF POOR SLEEP IS NOT A PERSONAL FAILING. IT IS A PUBLIC HEALTH CRISIS THAT DEMANDS ATTENTION AT EVERY LEVEL — INDIVIDUAL, MEDICAL, AND SOCIETAL. WHAT YOU WILL FIND IN THIS BOOK THE CHAPTERS THAT FOLLOW ARE ORGANIZED TO TAKE YOU FROM UNDERSTANDING TO ACTION. WE START WITH THE FUNDAMENTALS OF SLEEP SCIENCE SO YOU UNDERSTAND WHAT IS HAPPENING IN YOUR BODY AND BRAIN DURING THE NIGHT. WE THEN EXPLORE THE MAJOR SLEEP DISORDERS IN DETAIL — WHAT THEY ARE, HOW THEY PRESENT, AND HOW THEY ARE TREATED. FINALLY, WE MOVE INTO PRACTICAL STRATEGIES AND PERSONALIZED PLANS. EVERY CHAPTER INCLUDES RESEARCH-BASED INFORMATION PRESENTED IN ACCESSIBLE LANGUAGE. EVERY TREATMENT RECOMMENDATION IS GROUNDED IN THE SCIENTIFIC LITERATURE. EVERY PRACTICAL SUGGESTION HAS BEEN TESTED IN REAL LIFE BY REAL PEOPLE WHO HAVE NAVIGATED THESE CHALLENGES. A NOTE ON HOPE IF YOU HAVE STRUGGLED WITH SLEEP FOR YEARS, YOU MAY HAVE LOST HOPE. YOU MAY HAVE TRIED EVERYTHING — MEDICATIONS, SUPPLEMENTS, HERBAL TEAS, WHITE NOISE MACHINES, BLACKOUT CURTAINS, COUNTING SHEEP, MEDITATION APPS — AND NOTHING HAS WORKED. YOU MAY HAVE CONCLUDED THAT YOUR SLEEP IS SIMPLY BROKEN AND CANNOT BE FIXED. I WANT TO OFFER YOU A DIFFERENT PERSPECTIVE. SLEEP DISORDERS ARE MEDICAL CONDITIONS. LIKE OTHER MEDICAL CONDITIONS, THEY HAVE CAUSES, MECHANISMS, AND TREATMENTS. THE FACT THAT YOU HAVE NOT FOUND A SOLUTION YET DOES NOT MEAN NO SOLUTION EXISTS. IT MEANS YOU HAVE NOT YET FOUND

THE RIGHT SOLUTION FOR YOUR SPECIFIC SITUATION. THIS BOOK WILL HELP YOU FIND IT. IT WILL TEACH YOU HOW TO IDENTIFY WHAT IS GOING WRONG, HOW TO CHOOSE THE RIGHT TREATMENT, AND HOW TO IMPLEMENT IT IN A WAY THAT WORKS FOR YOUR LIFE. THE PATH IS NOT ALWAYS EASY, BUT IT IS ALWAYS WORTHWHILE.

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CHAPTER 1: THE SLEEP CRISIS — WHY YOU CAN'T SLEEP AND WHY IT MATTERS

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WE ARE IN THE MIDST OF A GLOBAL SLEEP CRISIS. ACROSS THE DEVELOPED WORLD, PEOPLE ARE SLEEPING LESS AND WORSE THAN AT ANY POINT IN MODERN HISTORY. THE NUMBERS ARE STAGGERING. ACCORDING TO THE CENTERS FOR DISEASE CONTROL AND PREVENTION, ONE IN THREE ADULTS DOES NOT GET ENOUGH SLEEP. THE WORLD HEALTH ORGANIZATION HAS CLASSIFIED SLEEP DEPRIVATION AS A PUBLIC HEALTH EPIDEMIC. AND THE PROBLEM IS GETTING WORSE. THE SCALE OF THE PROBLEM

INSOMNIA, THE MOST COMMON SLEEP DISORDER, AFFECTS APPROXIMATELY 30 PERCENT OF ADULTS AT SOME POINT IN THEIR LIVES. ABOUT 10 PERCENT OF ADULTS EXPERIENCE CHRONIC INSOMNIA, MEANING IT LASTS FOR THREE MONTHS OR LONGER. THAT IS ROUGHLY 800 MILLION PEOPLE WORLDWIDE WHO STRUGGLE TO FALL ASLEEP, STAY ASLEEP, OR FEEL RESTED AFTER SLEEP. SLEEP APNEA AFFECTS AN ESTIMATED 936 MILLION PEOPLE GLOBALLY, WITH 425 MILLION SUFFERING FROM MODERATE TO SEVERE FORMS. MOST OF THEM DO NOT KNOW THEY HAVE IT. RESTLESS LEGS SYNDROME AFFECTS BETWEEN 5 AND 10 PERCENT OF ADULTS. CIRCADIAN RHYTHM DISORDERS, INCLUDING THOSE CAUSED BY SHIFT WORK, AFFECT MILLIONS MORE. THESE ARE NOT NICHE CONDITIONS. THEY ARE WIDESPREAD, UNDERDIAGNOSED, AND UNDERTREATED. THE COST OF POOR SLEEP

THE CONSEQUENCES OF SLEEP DISORDERS EXTEND FAR BEYOND FEELING TIRED. CHRONIC SLEEP DEPRIVATION HAS BEEN LINKED TO

NEARLY EVERY MAJOR HEALTH CONDITION. THE LIST INCLUDES CARDIOVASCULAR DISEASE, HYPERTENSION, DIABETES, OBESITY, DEPRESSION, ANXIETY, WEAKENED IMMUNE FUNCTION, HORMONAL IMBALANCES, COGNITIVE DECLINE, AND AN INCREASED RISK OF ACCIDENTS. RESEARCH HAS FOUND THAT PEOPLE WHO SLEEP FEWER THAN SIX HOURS PER NIGHT HAVE A 48 PERCENT HIGHER RISK OF DEVELOPING OR DYING FROM HEART DISEASE. THE LINK BETWEEN POOR SLEEP AND TYPE 2 DIABETES IS SO STRONG THAT SLEEP QUALITY IS NOW CONSIDERED A MODIFIABLE RISK FACTOR ALONGSIDE DIET AND EXERCISE. SLEEP DEPRIVATION IMPAIRS IMMUNE FUNCTION WITHIN DAYS, MAKING YOU MORE SUSCEPTIBLE TO INFECTIONS. THE COGNITIVE EFFECTS ARE EQUALLY SERIOUS. AFTER SEVENTEEN HOURS OF WAKEFULNESS, COGNITIVE PERFORMANCE IS COMPARABLE TO HAVING A BLOOD ALCOHOL CONCENTRATION OF 0.05 PERCENT. AFTER TWENTY-FOUR HOURS, IT IS EQUIVALENT TO 0.10 PERCENT — ABOVE THE LEGAL DRIVING LIMIT IN MOST COUNTRIES. THIS IS WHY FATIGUED DRIVING CAUSES AN ESTIMATED 328,000 CRASHES PER YEAR IN THE UNITED STATES ALONE. THE ECONOMIC IMPACT THE ECONOMIC COST OF SLEEP DISORDERS IS ENORMOUS. LOST PRODUCTIVITY, WORKPLACE ACCIDENTS, HEALTHCARE EXPENSES, AND ABSENTEEISM RELATED TO POOR SLEEP COST DEVELOPED ECONOMIES HUNDREDS OF BILLIONS OF DOLLARS ANNUALLY. IN THE UNITED STATES ALONE, THE ESTIMATED ECONOMIC BURDEN OF SLEEP DEPRIVATION IS BETWEEN \$280 BILLION AND \$411 BILLION PER YEAR. BUT THE HUMAN COST IS WHAT MATTERS MOST. SLEEP DISORDERS STEAL YEARS OF LIFE, RELATIONSHIPS, AND WELL-BEING. THEY TURN VIBRANT PEOPLE INTO EXHAUSTED SHADOWS OF THEMSELVES. THEY CREATE A FOG THAT MAKES EVERYTHING HARDER — THINKING, FEELING, CONNECTING, FUNCTIONING. WHY SLEEP DISORDERS ARE

UNDERDIAGNOSED DESPITE THEIR PREVALENCE, SLEEP DISORDERS ARE DRAMATICALLY UNDERDIAGNOSED. STUDIES SUGGEST THAT UP TO 80 PERCENT OF PEOPLE WITH SLEEP APNEA HAVE NOT BEEN DIAGNOSED. THE AVERAGE PERSON WITH RESTLESS LEGS SYNDROME WAITS TEN YEARS FOR A CORRECT DIAGNOSIS. CHRONIC INSOMNIA OFTEN GOES UNTREATED BECAUSE PEOPLE BELIEVE IT IS NORMAL OR THAT NOTHING CAN BE DONE. THERE ARE SEVERAL REASONS FOR THIS. THE FIRST IS AWARENESS. MANY PEOPLE DO NOT KNOW WHAT CONSTITUTES A SLEEP DISORDER. THEY THINK THEIR SLEEP IS NORMAL WHEN IT IS NOT. THE SECOND IS NORMALIZATION. OUR CULTURE HAS NORMALIZED EXHAUSTION. BEING TIRED ALL THE TIME IS SEEN AS A BADGE OF HONOR RATHER THAN A SIGN THAT SOMETHING IS WRONG. THE THIRD IS ACCESS. SLEEP MEDICINE IS A SPECIALIZED FIELD, AND SLEEP STUDIES ARE EXPENSIVE AND SOMETIMES DIFFICULT TO ACCESS. THE SLEEP REVOLUTION THE GOOD NEWS IS THAT AWARENESS IS GROWING. THE PAST DECADE HAS SEEN AN EXPLOSION OF RESEARCH ON SLEEP. WE NOW UNDERSTAND THE MECHANISMS OF SLEEP BETTER THAN EVER BEFORE. WE HAVE EFFECTIVE TREATMENTS FOR MOST SLEEP DISORDERS. AND THE STIGMA AROUND SLEEP PROBLEMS IS SLOWLY DECREASING. THIS BOOK IS PART OF THAT REVOLUTION. IT IS BASED ON THE SCIENCE OF SLEEP MEDICINE, TRANSLATED INTO PRACTICAL KNOWLEDGE THAT ANYONE CAN USE. IT DRAWS ON COGNITIVE BEHAVIORAL THERAPY FOR INSOMNIA, OR CBT-I, WHICH IS THE GOLD STANDARD TREATMENT FOR CHRONIC INSOMNIA. IT COVERS THE LATEST RESEARCH ON SLEEP APNEA TREATMENTS, CIRCADIAN RHYTHM MANAGEMENT, AND THE CONNECTION BETWEEN SLEEP AND MENTAL HEALTH. THE FIRST STEP IF YOU ARE READING THIS BOOK, YOU HAVE ALREADY TAKEN THE FIRST STEP. YOU HAVE ACKNOWLEDGED THAT SLEEP MATTERS AND THAT YOU DESERVE

BETTER REST. THAT ACKNOWLEDGMENT IS POWERFUL. IT IS THE BEGINNING OF CHANGE. IN THE CHAPTERS THAT FOLLOW, WE WILL EXPLORE THE ARCHITECTURE OF SLEEP, THE MAJOR SLEEP DISORDERS, AND THE EVIDENCE-BASED STRATEGIES THAT CAN HELP YOU RECLAIM YOUR REST. BUT FIRST, LET US UNDERSTAND WHAT SLEEP ACTUALLY IS AND WHY YOUR BODY FIGHTS SO HARD TO GET IT.

REFLECTION QUESTIONS

On a scale of 1 to 10, how would you rate your sleep quality over the past month? When did you last feel completely rested after waking up? What is the biggest challenge you face when it comes to sleep? Have you ever discussed your sleep problems with a healthcare provider? If not, what stopped you?

The Sleep-Health Connection The relationship between sleep and physical health is bidirectional. Poor sleep increases the risk of health problems, and health problems make it harder to sleep. Understanding this connection is crucial for anyone trying to improve their sleep. One of the most well-documented connections is between sleep and cardiovascular health. Research has found that people who sleep fewer than six hours per night have a significantly higher risk of developing hypertension, coronary heart disease, and stroke. The mechanism appears to involve the autonomic nervous system. Sleep deprivation activates the sympathetic nervous system, increases heart rate, raises blood pressure, and promotes inflammation. The connection between sleep and metabolism is equally important. Sleep deprivation disrupts the hormones that regulate appetite — ghrelin increases, leptin decreases — leading to increased hunger and cravings for high-calorie foods. It also impairs

glucose metabolism, reducing insulin sensitivity by as much as 30 percent after just four nights of short sleep. This is why chronic sleep deprivation is a risk factor for obesity and type 2 diabetes.

Sleep and the Immune System

The immune system is profoundly affected by sleep. During deep sleep, the body produces cytokines — proteins that help fight infection and inflammation. Sleep deprivation reduces cytokine production and impairs the body's ability to mount an effective immune response. This has real-world consequences. People who sleep fewer than seven hours per night are nearly three times more likely to develop a cold when exposed to the virus compared to those who sleep eight hours or more. The effect of sleep on vaccine response is also well documented — people who sleep poorly after receiving a vaccine produce fewer antibodies and have a weaker immune response.

The Cognitive Toll

The cognitive effects of sleep deprivation are wide-ranging and serious. Attention is the first casualty. When you are sleep-deprived, your ability to sustain focus declines, and you are more easily distracted. Working memory — the ability to hold and manipulate information in your mind — is impaired. Decision-making becomes slower and less accurate. Memory consolidation occurs during sleep. During deep sleep, the brain replays and strengthens memories formed during the day, transferring them from short-term to long-term storage. During REM sleep, the brain integrates these memories with existing knowledge, creating new insights and creative connec

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CHAPTER 2: UNDERSTANDING YOUR SLEEP ARCHITECTURE

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BEFORE WE CAN FIX SLEEP, WE NEED TO UNDERSTAND IT. SLEEP IS NOT SIMPLY A STATE OF BEING UNCONSCIOUS. IT IS AN ACTIVE, DYNAMIC PROCESS INVOLVING DISTINCT STAGES THAT CYCLE THROUGHOUT THE NIGHT. UNDERSTANDING THIS ARCHITECTURE IS THE FOUNDATION FOR UNDERSTANDING WHAT GOES WRONG IN SLEEP DISORDERS. THE TWO-PROCESS MODEL SLEEP IS REGULATED BY TWO INTERACTING SYSTEMS: THE CIRCADIAN RHYTHM AND THE SLEEP HOMEOSTAT. THE CIRCADIAN RHYTHM IS YOUR INTERNAL BODY CLOCK, A ROUGHLY TWENTY-FOUR-HOUR CYCLE THAT REGULATES WHEN YOU FEEL ALERT AND WHEN YOU FEEL SLEEPY. IT IS DRIVEN BY THE SUPRACHIASMATIC NUCLEUS IN THE BRAIN, WHICH RESPONDS TO LIGHT CUES FROM THE ENVIRONMENT. THE SLEEP HOMEOSTAT, ON THE OTHER HAND, TRACKS HOW LONG YOU HAVE BEEN AWAKE. THE LONGER YOU STAY AWAKE, THE MORE SLEEP PRESSURE BUILDS UP, MAKING YOU INCREASINGLY TIRED. THINK OF IT LIKE A HUNGER DRIVE FOR SLEEP. THE LONGER YOU GO WITHOUT EATING, THE HUNGRIER YOU GET. THE LONGER YOU GO WITHOUT SLEEPING, THE SLEEPIER YOU GET. THESE TWO SYSTEMS WORK TOGETHER TO KEEP YOU AWAKE DURING THE DAY AND ASLEEP AT NIGHT. WHEN THEY FALL OUT OF SYNC, SLEEP DISORDERS CAN DEVELOP. THE STAGES OF SLEEP

SLEEP IS DIVIDED INTO TWO MAIN TYPES: NON-RAPID EYE MOVEMENT SLEEP, OR NREM, AND RAPID EYE MOVEMENT SLEEP, OR REM. NREM SLEEP HAS THREE STAGES. STAGE ONE IS THE

LIGHTEST STAGE OF SLEEP. IT LASTS ONLY A FEW MINUTES AND IS THE TRANSITION FROM WAKEFULNESS TO SLEEP. IN THIS STAGE, YOUR MUSCLES RELAX, YOUR HEARTBEAT SLOWS, AND YOUR BRAIN PRODUCES THETA WAVES. YOU CAN BE EASILY WOKEN FROM STAGE ONE, AND YOU MAY NOT EVEN REALIZE YOU WERE ASLEEP. STAGE TWO IS MORE STABLE. YOUR BRAIN PRODUCES BURSTS OF ACTIVITY CALLED SLEEP SPINDLES AND K-COMPLEXES, WHICH ARE BELIEVED TO PLAY A ROLE IN MEMORY CONSOLIDATION AND SENSORY PROCESSING. YOUR BODY TEMPERATURE DROPS, AND YOUR HEART RATE CONTINUES TO SLOW. YOU SPEND ABOUT 50 PERCENT OF YOUR TOTAL SLEEP TIME IN STAGE TWO. STAGE THREE IS DEEP SLEEP, ALSO CALLED SLOW-WAVE SLEEP. THIS IS THE MOST RESTORATIVE STAGE OF SLEEP. YOUR BRAIN PRODUCES DELTA WAVES, AND IT IS VERY DIFFICULT TO WAKE YOU. DURING DEEP SLEEP, YOUR BODY REPAIRS TISSUES, BUILDS BONE AND MUSCLE, STRENGTHENS THE IMMUNE SYSTEM, AND CLEARS METABOLIC WASTE FROM THE BRAIN. THIS IS THE STAGE THAT MAKES YOU FEEL REFRESHED IN THE MORNING. REM SLEEP, THE STAGE ASSOCIATED WITH DREAMING, OCCURS ABOUT NINETY MINUTES AFTER YOU FALL ASLEEP. YOUR EYES MOVE RAPIDLY BENEATH YOUR EYELIDS, YOUR BRAIN BECOMES NEARLY AS ACTIVE AS WHEN YOU ARE AWAKE, AND YOUR VOLUNTARY MUSCLES ARE TEMPORARILY PARALYZED TO PREVENT YOU FROM ACTING OUT YOUR DREAMS. REM SLEEP IS CRITICAL FOR EMOTIONAL REGULATION, MEMORY CONSOLIDATION, AND CREATIVE PROBLEM-SOLVING. THE SLEEP CYCLE THROUGHOUT THE NIGHT, YOU CYCLE THROUGH THESE STAGES APPROXIMATELY EVERY NINETY MINUTES. A HEALTHY SLEEPER WILL HAVE FOUR TO SIX COMPLETE CYCLES PER NIGHT. THE COMPOSITION OF THESE CYCLES CHANGES AS THE NIGHT PROGRESSES. IN THE FIRST HALF OF

THE NIGHT, THE CYCLES CONTAIN MORE DEEP SLEEP. THIS IS WHY WAKING SOMEONE EARLY IN THE NIGHT CAN LEAVE THEM FEELING PARTICULARLY GROGGY — THEY ARE BEING PULLED OUT OF DEEP SLEEP. IN THE SECOND HALF OF THE NIGHT, THE CYCLES CONTAIN MORE REM SLEEP. THIS IS WHY YOU ARE MORE LIKELY TO REMEMBER YOUR DREAMS IN THE MORNING AND WHY CUTTING YOUR SLEEP SHORT BY AN HOUR IN THE MORNING DISPROPORTIONATELY AFFECTS REM SLEEP. WHAT HAPPENS WHEN YOU DO NOT CYCLE PROPERLY SLEEP DISORDERS DISRUPT THESE CYCLES. INSOMNIA MAY PREVENT YOU FROM REACHING DEEP SLEEP OR STAYING IN IT LONG ENOUGH. SLEEP APNEA CAUSES REPEATED AWAKENINGS THAT FRAGMENT SLEEP AND PREVENT CYCLING. RESTLESS LEGS SYNDROME CAN MAKE IT IMPOSSIBLE TO FALL INTO DEEPER STAGES. THE RESULT IS NOT JUST LESS SLEEP, BUT LOWER QUALITY SLEEP THAT DOES NOT PROVIDE THE RESTORATIVE BENEFITS YOUR BODY NEEDS. THE GLYMPHATIC SYSTEM ONE OF THE MOST IMPORTANT DISCOVERIES ABOUT SLEEP IN RECENT YEARS IS THE GLYMPHATIC SYSTEM. THIS IS THE BRAIN'S WASTE CLEARANCE SYSTEM, WHICH BECOMES ACTIVE PRIMARILY DURING DEEP SLEEP. IT WORKS BY CIRCULATING CEREBROSPINAL FLUID THROUGH THE BRAIN, FLUSHING OUT METABOLIC WASTE PRODUCTS THAT ACCUMULATE DURING WAKING HOURS. ONE OF THE WASTES CLEARED BY THE GLYMPHATIC SYSTEM IS BETA-AMYLOID, A PROTEIN THAT FORMS THE PLAQUES ASSOCIATED WITH ALZHEIMER'S DISEASE. THIS DISCOVERY HAS LED TO A GROWING BODY OF RESEARCH LINKING CHRONIC SLEEP DEPRIVATION TO AN INCREASED RISK OF NEURODEGENERATIVE DISEASES. THE MESSAGE IS CLEAR: SLEEP IS NOT OPTIONAL FOR BRAIN HEALTH. IT IS WHEN YOUR BRAIN CLEANS ITSELF. WHY QUALITY MATTERS MORE THAN QUANTITY WHILE MOST GUIDELINES RECOMMEND SEVEN TO NINE HOURS OF SLEEP

FOR ADULTS, THE QUALITY OF THAT SLEEP MATTERS AT LEAST AS MUCH AS THE QUANTITY. SOMEONE WHO SLEEPS EIGHT HOURS BUT WAKES UP REPEATEDLY DURING THE NIGHT MAY FEEL WORSE THAN SOMEONE WHO SLEEPS SIX AND A HALF HOURS WITHOUT INTERRUPTION. SLEEP QUALITY IS DETERMINED BY HOW MUCH TIME YOU SPEND IN THE RESTORATIVE STAGES — DEEP SLEEP AND REM SLEEP — AND HOW SMOOTHLY YOU CYCLE THROUGH THE STAGES. IF YOUR SLEEP IS FRAGMENTED, THESE CYCLES ARE DISRUPTED, AND YOU DO NOT GET THE FULL BENEFIT OF YOUR TIME IN BED.

MEASURING YOUR SLEEP IF YOU ARE DEALING WITH A SLEEP DISORDER, IT CAN BE HELPFUL TO TRACK YOUR SLEEP FOR A FEW WEEKS. A SIMPLE SLEEP DIARY CAN REVEAL PATTERNS THAT YOU MIGHT NOT NOTICE OTHERWISE. RECORD WHEN YOU GO TO BED, WHEN YOU FALL ASLEEP, HOW OFTEN YOU WAKE UP, WHEN YOU WAKE UP FOR THE DAY, AND HOW RESTED YOU FEEL. SLEEP TRACKING DEVICES CAN PROVIDE ADDITIONAL DATA, BUT THEY ARE NOT MEDICAL DEVICES. THEY CAN ESTIMATE SLEEP STAGES AND DURATION, BUT THEY ARE LESS ACCURATE THAN LABORATORY POLYSOMNOGRAPHY. USE THEM AS GENERAL GUIDES, NOT DIAGNOSTIC TOOLS.

REFLECTION QUESTIONS

Which stage of sleep do you think you are missing the most? Why? Have you ever noticed that you feel more rested after a shorter night of uninterrupted sleep than a longer night of fragmented sleep? What time of day do you naturally feel most alert? What time do you naturally feel sleepest? If you could change one thing about your sleep cycle, what would it be?

The Role of Adenosine Adenosine is a neurotransmitter that plays a central role in sleep regulation. It accumulates in the brain throughout the day, binding to receptors that promote sleepiness. The longer you are awake, the more adenosine builds up, and the stronger your urge to sleep becomes. Caffeine works by blocking adenosine receptors. When caffeine binds to these receptors, adenosine cannot, so you do not feel the sleep pressure that has accumulated. But the adenosine does not go away. It continues to build up, and when the caffeine wears off, the accumulated adenosine floods the receptors, creating the familiar caffeine crash. Sleep clears adenosine from the brain. During deep sleep, the glymphatic system flushes out adenosine and other metabolic waste products. This is why you feel refreshed after a good night's sleep — your brain has been cleaned of the chemical signals of wakefulness.

Circadian Rhythms Beyond Sleep The circadian clock does not only regulate sleep. It influences nearly every biological process in the body. Body temperature, hormone release, digestion, immune function, and cognitive performance all follow circadian patterns. Cortisol, the primary stress hormone, follows a strong circadian rhythm. It peaks in the early morning, helping you wake up and feel alert, and declines throughout the day, reaching its lowest point in the middle of the night. Melatonin, the sleep hormone, follows the opposite pattern, rising in the evening and peaking in the middle of the night. Growth hormone is primarily released during deep sleep. This is why deep sleep is so important for tissue repair, muscle growth, and recovery from injury. Disrupting deep

sleep reduces growth hormone release and impairs the body's ability to repair itself. Individual Differences in Sleep
Not ever

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CHAPTER 3: INSOMNIA — THE LONG NIGHT

CHAPTER 3: INSOMNIA — THE LONG NIGHT

INSOMNIA IS THE MOST COMMON SLEEP DISORDER, AND IT IS ALSO THE ONE PEOPLE ARE MOST LIKELY TO MISUNDERSTAND. IT IS NOT SIMPLY DIFFICULTY FALLING ASLEEP. IT IS A PERSISTENT CONDITION THAT AFFECTS THE ABILITY TO INITIATE OR MAINTAIN SLEEP, OR TO GET RESTORATIVE SLEEP DESPITE HAVING ADEQUATE OPPORTUNITY TO SLEEP. THE THREE FACES OF INSOMNIA

INSOMNIA PRESENTS IN THREE PRIMARY FORMS. SLEEP ONSET INSOMNIA IS DIFFICULTY FALLING ASLEEP AT THE BEGINNING OF THE NIGHT. YOU LIE IN BED, TIRED BUT WIRED, AND THE MINUTES STRETCH INTO HOURS. SLEEP MAINTENANCE INSOMNIA IS DIFFICULTY STAYING ASLEEP. YOU FALL ASLEEP EASILY BUT WAKE UP IN THE MIDDLE OF THE NIGHT AND STRUGGLE TO GET BACK TO SLEEP. EARLY MORNING AWAKENING IS WAKING UP TOO EARLY AND BEING UNABLE TO FALL BACK ASLEEP, OFTEN FEELING UNREFRESHED. MANY PEOPLE EXPERIENCE A COMBINATION OF THESE FORMS. THE COMMON THREAD IS THAT THE SLEEP YOU DO GET IS NOT RESTORATIVE. YOU WAKE UP FEELING AS TIRED AS WHEN YOU WENT TO BED.

ACUTE VS. CHRONIC INSOMNIA

ACUTE INSOMNIA IS SHORT-TERM AND OFTEN TRIGGERED BY A SPECIFIC EVENT — STRESS AT WORK, A BREAKUP, TRAVEL ACROSS TIME ZONES, OR A MEDICAL PROCEDURE. IT TYPICALLY RESOLVES ON ITS OWN ONCE THE TRIGGER PASSES. ACUTE INSOMNIA CAN LAST FROM A FEW DAYS TO A FEW WEEKS. CHRONIC INSOMNIA, BY CONTRAST, LASTS FOR THREE MONTHS OR LONGER AND OCCURS AT LEAST THREE NIGHTS PER WEEK. IT OFTEN PERSISTS EVEN AFTER THE

ORIGINAL TRIGGER IS GONE. THIS IS BECAUSE INSOMNIA CAN BECOME SELF-PERPETUATING. THE FEAR OF NOT SLEEPING CREATES ANXIETY THAT PREVENTS SLEEP, WHICH REINFORCES THE FEAR, WHICH CREATES MORE ANXIETY, AND SO ON. THE HYPERAROUSAL MODEL THE LEADING THEORY OF CHRONIC INSOMNIA IS THE HYPERAROUSAL MODEL. THIS MODEL SUGGESTS THAT PEOPLE WITH INSOMNIA HAVE A NERVOUS SYSTEM THAT IS STUCK IN A STATE OF HEIGHTENED ALERTNESS. THEIR BODIES ARE IN A STATE OF PHYSIOLOGICAL HYPERAROUSAL, WITH ELEVATED HEART RATE, HIGHER CORTISOL LEVELS, AND INCREASED BRAIN ACTIVITY — EVEN DURING THE DAY. THIS HYPERAROUSAL MAKES IT DIFFICULT TO TRANSITION INTO SLEEP. THE BRAIN CANNOT SHUT OFF BECAUSE IT IS PRIMED FOR THREAT DETECTION. THIS IS WHY YOU CAN BE EXHAUSTED BUT STILL UNABLE TO SLEEP. YOUR BODY IS TIRED, BUT YOUR NERVOUS SYSTEM IS ON HIGH ALERT. THE VICIOUS CYCLE OF INSOMNIA INSOMNIA CREATES A VICIOUS CYCLE THAT IS HARD TO BREAK. IT STARTS WITH A BAD NIGHT OF SLEEP. THE NEXT DAY, YOU FEEL TIRED AND IRRITABLE. YOU START WORRYING ABOUT WHETHER YOU WILL SLEEP THAT NIGHT. AT BEDTIME, YOU ARE ANXIOUS. YOU TRY HARDER TO FALL ASLEEP, BUT TRYING HARDER TO SLEEP IS LIKE TRYING HARDER TO DIGEST FOOD — IT DOES NOT WORK. THE EFFORT ITSELF KEEPS YOU AWAKE. OVER TIME, YOU DEVELOP NEGATIVE ASSOCIATIONS WITH YOUR BED. YOUR BED BECOMES A PLACE OF FRUSTRATION RATHER THAN REST. YOU START SPENDING MORE TIME IN BED, HOPING TO CATCH UP ON SLEEP, BUT THIS ONLY FRAGMENTS YOUR SLEEP FURTHER. YOU START NAPPING DURING THE DAY, WHICH REDUCES YOUR SLEEP DRIVE AT NIGHT. THE CYCLE TIGHTENS. THE ROLE OF RACING THOUGHTS ONE OF THE MOST COMMON COMPLAINTS AMONG PEOPLE WITH INSOMNIA IS RACING

THOUGHTS. AS SOON AS THE HEAD HITS THE PILLOW, THE MIND STARTS CHURNING — REPLAYING CONVERSATIONS, PLANNING TOMORROW, WORRYING ABOUT FINANCES, CREATING WORST-CASE SCENARIOS. THIS IS NOT RANDOM NOISE. IT IS A SYMPTOM OF THE HYPERAROUSAL STATE. THE BRAIN IS PROCESSING INFORMATION AND SCANNING FOR THREATS. IN THE QUIET OF THE BEDROOM, WITH NO EXTERNAL DISTRACTIONS, THESE THOUGHTS BECOME LOUDER. THE MORE YOU TRY TO SUPPRESS THEM, THE MORE PERSISTENT THEY BECOME. COGNITIVE BEHAVIORAL THERAPY FOR INSOMNIA THE GOLD STANDARD TREATMENT FOR CHRONIC INSOMNIA IS COGNITIVE BEHAVIORAL THERAPY FOR INSOMNIA, OR CBT-I. IT IS A STRUCTURED PROGRAM THAT ADDRESSES THE THOUGHTS, BEHAVIORS, AND EMOTIONS THAT PERPETUATE INSOMNIA. UNLIKE SLEEP MEDICATIONS, WHICH PROVIDE TEMPORARY RELIEF, CBT-I ADDRESSES THE UNDERLYING CAUSES AND PRODUCES LASTING IMPROVEMENT. CBT-I INCLUDES SEVERAL COMPONENTS. SLEEP RESTRICTION THERAPY CONSOLIDATES YOUR SLEEP BY LIMITING THE TIME YOU SPEND IN BED TO MATCH YOUR ACTUAL SLEEP TIME. THIS INCREASES SLEEP DRIVE AND REDUCES TIME SPENT LYING AWAKE. STIMULUS CONTROL THERAPY REBUILDS THE ASSOCIATION BETWEEN YOUR BED AND SLEEP BY INSTRUCTING YOU TO GET OUT OF BED WHEN YOU CANNOT SLEEP AND ONLY RETURN WHEN YOU FEEL SLEEPY. COGNITIVE THERAPY ADDRESSES THE UNHELPFUL BELIEFS AND WORRIES ABOUT SLEEP THAT FUEL INSOMNIA. RELAXATION TECHNIQUES REDUCE PHYSIOLOGICAL AROUSAL BEFORE BED. RESEARCH HAS SHOWN THAT CBT-I IS AS EFFECTIVE AS SLEEP MEDICATIONS IN THE SHORT TERM AND MORE EFFECTIVE IN THE LONG TERM. IT IS RECOMMENDED BY THE AMERICAN COLLEGE OF PHYSICIANS, THE BRITISH ASSOCIATION FOR PSYCHOPHARMACOLOGY, AND THE EUROPEAN SLEEP

RESEARCH SOCIETY AS THE FIRST-LINE TREATMENT FOR CHRONIC INSOMNIA. SLEEP MEDICATIONS: THE PROS AND CONS SLEEP MEDICATIONS CAN BE HELPFUL IN CERTAIN SITUATIONS, BUT THEY ARE NOT A LONG-TERM SOLUTION FOR MOST PEOPLE. THE MOST COMMONLY PRESCRIBED SLEEP MEDICATIONS ARE BENZODIAZEPINE RECEPTOR AGONISTS, WHICH INCLUDE DRUGS LIKE ZOLPIDEM AND ESZOPICLONE. THEY WORK BY ENHANCING THE EFFECT OF GABA, A NEUROTRANSMITTER THAT INHIBITS BRAIN ACTIVITY. THESE MEDICATIONS CAN HELP WITH SHORT-TERM INSOMNIA, PARTICULARLY WHEN USED FOR A FEW DAYS TO A FEW WEEKS. BUT THEY COME WITH SIGNIFICANT RISKS. TOLERANCE DEVELOPS QUICKLY, MEANING YOU NEED HIGHER DOSES TO GET THE SAME EFFECT. DEPENDENCE IS COMMON, AND WITHDRAWAL CAN CAUSE REBOUND INSOMNIA THAT IS WORSE THAN THE ORIGINAL PROBLEM. SIDE EFFECTS INCLUDE DAYTIME DROWSINESS, MEMORY IMPAIRMENT, AND AN INCREASED RISK OF FALLS IN OLDER ADULTS. MELATONIN SUPPLEMENTS ARE A GENTLER OPTION, BUT THEY ARE NOT A SEDATIVE. MELATONIN IS A HORMONE THAT SIGNALS TO YOUR BODY THAT IT IS TIME TO SLEEP. IT IS MOST EFFECTIVE FOR CIRCADIAN RHYTHM DISORDERS AND JET LAG, NOT FOR CHRONIC INSOMNIA. PRACTICAL STEPS FOR MANAGING INSOMNIA IF YOU ARE STRUGGLING WITH INSOMNIA, THERE ARE SEVERAL THINGS YOU CAN START DOING TODAY. FIRST, ESTABLISH A CONSISTENT SLEEP SCHEDULE, EVEN ON WEEKENDS. YOUR BRAIN CRAVES ROUTINE. SECOND, CREATE A WIND-DOWN ROUTINE THAT SIGNALS TO YOUR BODY THAT SLEEP IS COMING. THIS MIGHT INCLUDE DIMMING LIGHTS, READING A PHYSICAL BOOK, GENTLE STRETCHING, OR A WARM BATH. THIRD, KEEP YOUR BEDROOM COOL, DARK, AND QUIET. FOURTH, GET OUT OF BED IF YOU CANNOT SLEEP AFTER TWENTY

MINUTES. LYING IN BED AWAKE TRAINS YOUR BRAIN TO ASSOCIATE BED WITH WAKEFULNESS. FIFTH, LIMIT CAFFEINE AND ALCOHOL, ESPECIALLY IN THE EVENING. REFLECTION QUESTIONS

When did your insomnia start? What was happening in your life at that time? What thoughts go through your mind when you are lying in bed unable to sleep? Have you ever tried sleep medications? What was your experience? What is one small change you could make tonight to improve your sleep environment?

The Role of Genetics in Insomnia Research has found that insomnia has a significant genetic component. Twin studies estimate that the heritability of insomnia is about 40 to 50 percent, meaning that roughly half of the variation in insomnia risk between people can be attributed to genetic factors. Several genetic variants have been identified that increase the risk of insomnia. Some of these variants affect neurotransmitter systems involved in sleep regulation, including the GABA and dopamine systems. Others affect the circadian clock genes that regulate sleep timing. The genetic link explains why insomnia runs in families. It also means that some people are inherently more vulnerable to insomnia than others. This is not a character flaw. It is a biological predisposition, like having a family history of high blood pressure or diabetes. Insomnia in Different Populations Insomnia does not affect all populations equally. Women are significantly more likely than men to experience insomnia. The gender difference emerges during puberty and persists throughout life. Hormonal fluctuations related to the menstrual cycle, p

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CHAPTER 4: SLEEP APNEA AND BREATHING DISORDERS

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SLEEP APNEA IS ONE OF THE MOST COMMON AND MOST DANGEROUS SLEEP DISORDERS. IT IS ALSO ONE OF THE MOST TREATABLE. YET THE VAST MAJORITY OF PEOPLE WHO HAVE IT DO NOT KNOW IT. UNDERSTANDING SLEEP APNEA COULD SAVE YOUR LIFE — AND IT MIGHT IMPROVE THE QUALITY OF YOUR SLEEP MORE THAN ANYTHING ELSE YOU DO. WHAT IS SLEEP APNEA? SLEEP APNEA IS A CONDITION IN WHICH BREATHING REPEATEDLY STOPS AND STARTS DURING SLEEP. THESE PAUSES, CALLED APNEAS, CAN LAST FROM TEN SECONDS TO OVER A MINUTE. THEY CAN OCCUR HUNDREDS OF TIMES PER NIGHT, EACH ONE WAKING YOU BRIEFLY — OFTEN WITHOUT YOUR AWARENESS — AS YOUR BRAIN FIGHTS TO RESTART BREATHING. THE MOST COMMON FORM IS OBSTRUCTIVE SLEEP APNEA, OR OSA, WHICH OCCURS WHEN THE MUSCLES IN THE BACK OF THE THROAT RELAX EXCESSIVELY DURING SLEEP. THIS RELAXATION CAUSES THE SOFT TISSUE AT THE BACK OF THE THROAT TO COLLAPSE AND BLOCK THE AIRWAY. WHEN YOUR BRAIN SENSES THAT YOU ARE NOT GETTING ENOUGH OXYGEN, IT SENDS A SIGNAL TO WAKE YOU JUST ENOUGH TO REOPEN THE AIRWAY. YOU MIGHT GASP, SNORT, OR MAKE A CHOKING SOUND AS YOU RESUME BREATHING. CENTRAL SLEEP APNEA, OR CSA, IS LESS COMMON. IT OCCURS WHEN THE BRAIN FAILS TO SEND THE PROPER SIGNALS TO THE MUSCLES THAT CONTROL BREATHING. UNLIKE OSA, WHERE THERE IS A PHYSICAL BLOCKAGE, CSA IS A FAILURE OF THE RESPIRATORY CONTROL CENTER IN THE BRAIN. COMPLEX SLEEP APNEA, ALSO CALLED

TREATMENT-EMERGENT CENTRAL SLEEP APNEA, IS A COMBINATION OF BOTH TYPES. THE SYMPTOMS YOU MIGHT MISS THE CLASSIC SYMPTOM OF SLEEP APNEA IS LOUD, DISRUPTIVE SNORING FOLLOWED BY EPISODES OF SILENCE, THEN A GASP OR SNORT. BUT NOT EVERYONE WHO SNORES HAS SLEEP APNEA, AND NOT EVERYONE WITH SLEEP APNEA SNORES. OTHER SYMPTOMS INCLUDE EXCESSIVE DAYTIME SLEEPINESS, WAKING UP WITH A DRY MOUTH OR SORE THROAT, MORNING HEADACHES, DIFFICULTY CONCENTRATING, IRRITABILITY, AND WAKING UP FREQUENTLY DURING THE NIGHT TO URINATE. MANY PEOPLE WITH SLEEP APNEA DO NOT REALIZE THEY HAVE IT BECAUSE THEY DO NOT REMEMBER WAKING UP. EACH APNEA EVENT CAUSES A MICRO-AROUSAL — A BRIEF AWAKENING THAT MIGHT LAST ONLY A FEW SECONDS. THESE MICRO-AROUSALS DO NOT REGISTER IN CONSCIOUS MEMORY, BUT THEY PREVENT THE BRAIN FROM ENTERING OR MAINTAINING DEEP SLEEP. THE HEALTH CONSEQUENCES UNTREATED SLEEP APNEA IS LINKED TO A LONG LIST OF SERIOUS HEALTH CONDITIONS. THE REPEATED DROPS IN BLOOD OXYGEN LEVELS PUT STRESS ON THE CARDIOVASCULAR SYSTEM. RESEARCH HAS FOUND THAT UNTREATED SLEEP APNEA INCREASES THE RISK OF HIGH BLOOD PRESSURE, HEART ATTACK, STROKE, ATRIAL FIBRILLATION, AND HEART FAILURE. THE RISK IS SIGNIFICANT — PEOPLE WITH SEVERE UNTREATED SLEEP APNEA HAVE A TWO TO THREE TIMES HIGHER RISK OF STROKE. SLEEP APNEA IS ALSO STRONGLY LINKED TO TYPE 2 DIABETES. THE INTERMITTENT HYPOXIA AND SLEEP FRAGMENTATION CAUSED BY APNEA DISRUPT GLUCOSE METABOLISM AND INCREASE INSULIN RESISTANCE. TREATING SLEEP APNEA HAS BEEN SHOWN TO IMPROVE BLOOD SUGAR CONTROL IN PEOPLE WITH DIABETES. BEYOND THE PHYSICAL EFFECTS, SLEEP APNEA TAKES A HEAVY TOLL ON COGNITIVE

FUNCTION AND MENTAL HEALTH. THE CHRONIC SLEEP DEPRIVATION CAUSED BY FRAGMENTED SLEEP IMPAIRS MEMORY, ATTENTION, AND DECISION-MAKING. IT CONTRIBUTES TO DEPRESSION AND ANXIETY. IT INCREASES THE RISK OF MOTOR VEHICLE ACCIDENTS BY TWO TO THREE TIMES. WHO IS AT RISK CERTAIN FACTORS INCREASE THE RISK OF SLEEP APNEA. EXCESS WEIGHT IS THE STRONGEST RISK FACTOR, PARTICULARLY FAT DEPOSITS AROUND THE UPPER AIRWAY. HOWEVER, SLEEP APNEA ALSO AFFECTS PEOPLE WHO ARE NOT OVERWEIGHT. MEN ARE TWO TO THREE TIMES MORE LIKELY TO HAVE SLEEP APNEA THAN WOMEN, THOUGH THE RISK FOR WOMEN INCREASES AFTER MENOPAUSE. AGE IS A FACTOR — THE RISK INCREASES WITH AGE. FAMILY HISTORY, SMOKING, NASAL CONGESTION, AND THE USE OF ALCOHOL OR SEDATIVES BEFORE BED ALSO INCREASE THE RISK. DIAGNOSIS THE GOLD STANDARD FOR DIAGNOSING SLEEP APNEA IS A POLYSOMNOGRAM, OR SLEEP STUDY. THIS TEST RECORDS BRAIN ACTIVITY, EYE MOVEMENTS, HEART RATE, BREATHING PATTERNS, BLOOD OXYGEN LEVELS, AND BODY MOVEMENTS DURING SLEEP. SLEEP STUDIES CAN BE DONE IN A SLEEP LABORATORY OR AT HOME USING A PORTABLE MONITORING DEVICE. THE SEVERITY OF SLEEP APNEA IS MEASURED BY THE APNEA-HYPOPNEA INDEX, OR AHI, WHICH COUNTS THE NUMBER OF APNEA AND HYPOPNEA EVENTS PER HOUR OF SLEEP. AN AHI OF FIVE TO FIFTEEN IS CONSIDERED MILD, FIFTEEN TO THIRTY IS MODERATE, AND OVER THIRTY IS SEVERE. TREATMENT OPTIONS CONTINUOUS POSITIVE AIRWAY PRESSURE, OR CPAP, IS THE MOST COMMON AND EFFECTIVE TREATMENT FOR MODERATE TO SEVERE OSA. A CPAP MACHINE DELIVERS A STEADY STREAM OF PRESSURIZED AIR THROUGH A MASK THAT YOU WEAR OVER YOUR NOSE, OR BOTH NOSE AND MOUTH, WHILE YOU SLEEP. THE AIR PRESSURE KEEPS YOUR AIRWAY OPEN, PREVENTING COLLAPSES AND

ALLOWING YOU TO BREATHE NORMALLY THROUGHOUT THE NIGHT. CPAP IS HIGHLY EFFECTIVE WHEN USED CONSISTENTLY. IT ELIMINATES APNEA EVENTS, RESTORES NORMAL OXYGEN LEVELS, AND ALLOWS YOU TO ENTER DEEP SLEEP. MOST PEOPLE REPORT FEELING SIGNIFICANTLY BETTER WITHIN THE FIRST FEW NIGHTS OF TREATMENT. HOWEVER, CPAP CAN TAKE SOME GETTING USED TO. THE MASK CAN FEEL UNCOMFORTABLE, AND THE NOISE OF THE MACHINE CAN BE INTRUSIVE. WORKING WITH A SLEEP SPECIALIST TO FIND THE RIGHT MASK AND PRESSURE SETTINGS MAKES A SIGNIFICANT DIFFERENCE. OTHER TREATMENT OPTIONS INCLUDE ORAL APPLIANCES THAT REPOSITION THE JAW TO KEEP THE AIRWAY OPEN, POSITIONAL THERAPY THAT ENCOURAGES SLEEPING ON YOUR SIDE RATHER THAN YOUR BACK, AND IN SOME CASES, SURGERY TO REMOVE EXCESS TISSUE FROM THE THROAT. FOR PEOPLE WHO CANNOT TOLERATE CPAP, THESE ALTERNATIVES CAN BE EFFECTIVE. LIFESTYLE CHANGES THAT HELP CERTAIN LIFESTYLE CHANGES CAN REDUCE THE SEVERITY OF SLEEP APNEA. WEIGHT LOSS IS THE MOST EFFECTIVE — EVEN A 10 PERCENT REDUCTION IN BODY WEIGHT CAN SIGNIFICANTLY IMPROVE THE AHI. AVOIDING ALCOHOL BEFORE BED IS IMPORTANT BECAUSE ALCOHOL RELAXES THE THROAT MUSCLES AND WORSENS APNEA. SLEEPING ON YOUR SIDE RATHER THAN YOUR BACK CAN HELP. QUITTING SMOKING REDUCES INFLAMMATION IN THE AIRWAY. WHEN TO SEE A DOCTOR IF YOU HAVE BEEN TOLD THAT YOU SNORE LOUDLY, IF YOU WAKE UP GASPING OR CHOKING, IF YOU ARE EXCESSIVELY SLEEPY DURING THE DAY DESPITE WHAT SEEMS LIKE ADEQUATE SLEEP, OR IF YOU HAVE RISK FACTORS FOR SLEEP APNEA, IT IS WORTH DISCUSSING WITH A HEALTHCARE PROVIDER. SLEEP APNEA IS A MEDICAL CONDITION THAT REQUIRES

PROFESSIONAL DIAGNOSIS AND TREATMENT. IGNORING IT DOES NOT MAKE IT GO AWAY — IT ALLOWS THE DAMAGE TO ACCUMULATE.
REFLECTION QUESTIONS

Do you snore? Has anyone ever told you that you stop breathing during sleep? How sleepy are you during the day? Do you fall asleep easily in situations where you should be alert — reading, watching TV, driving? Have you ever woken up with a headache, dry mouth, or a feeling of choking? If you have been diagnosed with sleep apnea, what has been your experience with treatment?

The Mask of Sleep Apnea Sleep apnea is sometimes called the silent thief of sleep because it often goes undetected for years. Many people with sleep apnea dismiss their symptoms as normal aging, poor fitness, or just being a naturally tired person. They do not realize that their exhaustion has a specific, treatable cause. One reason sleep apnea is so often missed is that the person experiencing it does not know it is happening. The micro-arousals that occur after each apnea event are brief enough that they do not register in conscious awareness. The person may genuinely believe they slept through the night, even though their brain was fighting to breathe every few minutes. Bed partners are often the first to notice the signs. If your partner has told you that you snore loudly, stop breathing during sleep, or make gasping sounds, take it seriously. This is not just about noise — it is a medical red flag. Children and Sleep Apnea Sleep apnea also affects children, and it often presents differently than in adults. Instead of excessive daytime sleepiness, children with sleep apnea may show hyperactivity, inattention, irr

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CHAPTER 5: RESTLESS LEGS, RESTLESS NIGHTS — MOVEMENT DISORDERS

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IMAGINE THIS: YOU ARE EXHAUSTED. YOUR BODY ACHES FOR SLEEP. YOU LIE DOWN, CLOSE YOUR EYES, AND JUST AS YOU BEGIN TO DRIFT OFF, AN UNBEARABLE SENSATION CREEPS INTO YOUR LEGS. IT IS NOT QUITE PAIN AND NOT QUITE ITCHING. IT IS A DEEP, CRAWLING, PULLING FEELING THAT CANNOT BE IGNORED. THE ONLY RELIEF IS TO MOVE. SO YOU MOVE. AND THE CYCLE STARTS AGAIN. THIS IS RESTLESS LEGS SYNDROME, OR RLS, A NEUROLOGICAL DISORDER THAT AFFECTS BETWEEN 5 AND 10 PERCENT OF ADULTS. IT IS ONE OF THE MOST COMMON CAUSES OF SLEEP DISRUPTION, AND ONE OF THE MOST MISUNDERSTOOD. WHAT IS RESTLESS LEGS SYNDROME? RESTLESS LEGS SYNDROME IS CHARACTERIZED BY AN IRRESISTIBLE URGE TO MOVE THE LEGS, USUALLY ACCOMPANIED BY UNCOMFORTABLE SENSATIONS. THESE SENSATIONS ARE OFTEN DESCRIBED AS CREEPING, CRAWLING, PULLING, TINGLING, ACHING, OR ELECTRICAL. THEY TYPICALLY OCCUR WHEN THE PERSON IS AT REST — SITTING OR LYING DOWN — AND ARE WORST IN THE EVENING AND AT NIGHT. MOVEMENT PROVIDES TEMPORARY RELIEF, BUT THE SENSATIONS RETURN AS SOON AS YOU STOP MOVING. THE SEVERITY OF RLS VARIES WIDELY. SOME PEOPLE EXPERIENCE MILD SYMPTOMS A FEW TIMES PER MONTH. OTHERS EXPERIENCE SYMPTOMS EVERY NIGHT, MULTIPLE TIMES PER NIGHT, MAKING SLEEP NEARLY IMPOSSIBLE. THE CONDITION TENDS TO WORSEN WITH AGE AND CAN BECOME PROGRESSIVELY MORE SEVERE. THE LINK TO PERIODIC LIMB

MOVEMENT DISORDER ABOUT 80 TO 90 PERCENT OF PEOPLE WITH RLS ALSO HAVE PERIODIC LIMB MOVEMENT DISORDER, OR PLMD. PLMD INVOLVES REPETITIVE JERKING OR TWITCHING OF THE LEGS DURING SLEEP. THESE MOVEMENTS OCCUR EVERY TWENTY TO FORTY SECONDS AND CAN LAST FOR MINUTES OR HOURS. LIKE THE MICRO-AROUSALS IN SLEEP APNEA, THESE MOVEMENTS OFTEN DO NOT WAKE THE PERSON FULLY, BUT THEY FRAGMENT SLEEP AND PREVENT DEEP RESTORATIVE STAGES. PLMD CAN ALSO OCCUR IN PEOPLE WHO DO NOT HAVE RLS. IT IS MORE COMMON IN OLDER ADULTS AND CAN BE A SIDE EFFECT OF CERTAIN MEDICATIONS, PARTICULARLY ANTIDEPRESSANTS. WHAT CAUSES RLS? THE EXACT CAUSE OF RLS IS NOT FULLY UNDERSTOOD, BUT RESEARCH POINTS TO SEVERAL FACTORS. THE STRONGEST EVIDENCE IMPLICATES DYSFUNCTION IN THE BRAIN'S DOPAMINE PATHWAYS. DOPAMINE IS A NEUROTRANSMITTER INVOLVED IN CONTROLLING MOVEMENT. IN PEOPLE WITH RLS, THE BRAIN'S DOPAMINE SIGNALING APPEARS TO BE DISRUPTED, PARTICULARLY IN THE EVENING AND AT NIGHT. IRON DEFICIENCY IS CLOSELY LINKED TO RLS. IRON IS A COFACTOR IN DOPAMINE PRODUCTION, AND LOW IRON LEVELS IN THE BRAIN CAN DISRUPT DOPAMINE SIGNALING. MANY PEOPLE WITH RLS HAVE NORMAL BLOOD IRON LEVELS BUT LOW IRON LEVELS IN THE BRAIN. THIS IS WHY IRON SUPPLEMENTATION CAN HELP SOME PEOPLE WITH RLS, EVEN WHEN THEIR BLOOD TESTS SHOW NORMAL IRON LEVELS. GENETICS ALSO PLAY A ROLE. RLS TENDS TO RUN IN FAMILIES, AND SEVERAL GENETIC VARIANTS HAVE BEEN LINKED TO THE CONDITION. IF YOU HAVE A FIRST-DEGREE RELATIVE WITH RLS, YOUR RISK IS THREE TO SIX TIMES HIGHER THAN THE GENERAL POPULATION. OTHER FACTORS THAT CAN TRIGGER OR WORSEN RLS INCLUDE PREGNANCY, KIDNEY FAILURE, PERIPHERAL NEUROPATHY, AND

CERTAIN MEDICATIONS — INCLUDING SOME ANTIDEPRESSANTS, ANTIHISTAMINES, AND ANTIPSYCHOTICS. THE IMPACT ON SLEEP RLS CAN DEVASTATE SLEEP. THE SYMPTOMS MAKE IT DIFFICULT TO FALL ASLEEP, AND THEY OFTEN RETURN DURING THE NIGHT, CAUSING FREQUENT AWAKENINGS. THE LEG MOVEMENTS OF PLMD FURTHER FRAGMENT SLEEP. THE RESULT IS CHRONIC SLEEP DEPRIVATION THAT AFFECTS EVERY ASPECT OF LIFE. PEOPLE WITH RLS OFTEN REPORT SEVERE DAYTIME FATIGUE, DIFFICULTY CONCENTRATING, IRRITABILITY, AND DEPRESSED MOOD. THE CONDITION CAN INTERFERE WITH WORK, RELATIONSHIPS, AND DAILY ACTIVITIES. MANY PEOPLE WITH RLS REPORT THAT THE CONDITION SIGNIFICANTLY IMPAIRS THEIR QUALITY OF LIFE. DIAGNOSIS RLS IS DIAGNOSED BASED ON CLINICAL CRITERIA. THE ESSENTIAL CRITERIA INCLUDE AN URGE TO MOVE THE LEGS, USUALLY ACCOMPANIED BY UNCOMFORTABLE SENSATIONS; SYMPTOMS THAT BEGIN OR WORSEN DURING PERIODS OF REST OR INACTIVITY; SYMPTOMS THAT ARE PARTIALLY OR TOTALLY RELIEVED BY MOVEMENT; SYMPTOMS THAT OCCUR OR WORSEN IN THE EVENING OR AT NIGHT; AND THE EXCLUSION OF OTHER CONDITIONS THAT COULD EXPLAIN THE SYMPTOMS. THERE IS NO DEFINITIVE DIAGNOSTIC TEST FOR RLS, THOUGH A SLEEP STUDY CAN DETECT PLMD AND RULE OUT OTHER SLEEP DISORDERS. BLOOD TESTS MAY BE DONE TO CHECK IRON LEVELS, KIDNEY FUNCTION, AND OTHER POTENTIAL CAUSES.

TREATMENT OPTIONS TREATMENT FOR RLS DEPENDS ON THE SEVERITY OF SYMPTOMS. FOR MILD TO MODERATE RLS, LIFESTYLE CHANGES CAN BE EFFECTIVE. REDUCING OR ELIMINATING CAFFEINE, ALCOHOL, AND NICOTINE, PARTICULARLY IN THE EVENING, CAN HELP. REGULAR MODERATE EXERCISE, SUCH AS WALKING OR SWIMMING, CAN REDUCE SYMPTOMS, THOUGH INTENSE

EXERCISE CLOSE TO BEDTIME MAY WORSEN THEM. STRETCHING THE LEGS BEFORE BED, TAKING A WARM BATH, OR USING HEAT OR COLD PACKS ON THE LEGS CAN PROVIDE TEMPORARY RELIEF. FOR MODERATE TO SEVERE RLS, MEDICATIONS MAY BE NEEDED. THE MOST COMMONLY PRESCRIBED MEDICATIONS ARE DOPAMINE AGONISTS LIKE PRAMIPEXOLE AND ROPINIROLE. THESE DRUGS MIMIC DOPAMINE IN THE BRAIN AND CAN SIGNIFICANTLY REDUCE RLS SYMPTOMS. HOWEVER, THEY COME WITH SIGNIFICANT RISKS. LONG-TERM USE CAN LEAD TO AUGMENTATION — A PARADOXICAL WORSENING OF SYMPTOMS IN WHICH THEY OCCUR EARLIER IN THE DAY, SPREAD TO OTHER PARTS OF THE BODY, AND BECOME MORE INTENSE. AUGMENTATION AFFECTS A SUBSTANTIAL PERCENTAGE OF PEOPLE WHO USE DOPAMINE AGONISTS LONG-TERM. ALPHA-2-DELTA LIGANDS, SUCH AS GABAPENTIN AND PREGABALIN, ARE INCREASINGLY USED AS FIRST-LINE TREATMENTS. THEY ARE EFFECTIVE FOR RLS AND HAVE A LOWER RISK OF AUGMENTATION. FOR PEOPLE WITH LOW IRON LEVELS, IRON SUPPLEMENTATION CAN BE EFFECTIVE. SEVERE RLS THAT DOES NOT RESPOND TO OTHER TREATMENTS MAY BE TREATED WITH OPIOIDS, THOUGH THESE ARE USED CAUTIOUSLY DUE TO THE RISK OF DEPENDENCE.

PRACTICAL STRATEGIES FOR MANAGING RLS

IF YOU HAVE RLS, THERE ARE SEVERAL STRATEGIES YOU CAN USE TO MANAGE SYMPTOMS.

ESTABLISH A CONSISTENT SLEEP SCHEDULE. CREATE A RELAXING BEDTIME ROUTINE THAT INCLUDES GENTLE LEG STRETCHES OR MASSAGE. AVOID CAFFEINE AND ALCOHOL IN THE EVENING. EXERCISE REGULARLY, BUT NOT RIGHT BEFORE BED. CONSIDER USING A WARM BATH OR HEATING PAD ON YOUR LEGS BEFORE SLEEP. IF YOU HAVE PLMD, MAKE SURE YOUR BEDROOM IS COOL, AS OVERHEATING CAN WORSEN SYMPTOMS.

REFLECTION QUESTIONS

Have you ever experienced uncomfortable sensations in your legs that made it difficult to sleep? Do you notice that your symptoms are worse when you are tired, stressed, or have had caffeine? Have you had your iron levels checked recently? If so, what were the results? What activities or strategies have you found that provide temporary relief from your symptoms?

Restless Legs Syndrome and Pregnancy Pregnancy is a common trigger for restless legs syndrome. Between 20 and 30 percent of pregnant women experience RLS, particularly in the third trimester. The symptoms typically resolve within a few weeks of giving birth, but they can be very disruptive during pregnancy. The cause of pregnancy-related RLS is not fully understood, but several factors may contribute. Iron deficiency is common in pregnancy and is a known trigger for RLS. Hormonal changes, particularly increased estrogen and progesterone, may also play a role. The increased blood volume and metabolic demands of pregnancy can further stress the body's systems. Treatment options for RLS during pregnancy are limited because many RLS medications are not recommended during pregnancy. Iron supplementation, if iron levels are low, is safe and can be effective. Lifestyle modifications, such as moderate exercise, leg stretches, warm baths, and avoiding caffeine, are the primary approaches. Most women find relief knowing that the condition is temporary and will resolve after delivery. RLS in Children Restless legs syndrome also affects children, though it is often misdiagnosed. Children with RLS may not have the vocabulary to describe the sensations in their legs. Instead, they may complain of growing pain

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CHAPTER 6: CIRCADIAN RHYTHM DISORDERS — WHEN YOUR BODY CLOCK BREAKS

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YOUR BODY HAS AN INTERNAL CLOCK. IT IS LOCATED IN THE SUPRACHIASMATIC NUCLEUS OF THE HYPOTHALAMUS, AND IT REGULATES NEARLY EVERY BIOLOGICAL PROCESS IN YOUR BODY. IT TELLS YOU WHEN TO WAKE UP, WHEN TO EAT, WHEN TO RELEASE HORMONES, AND WHEN TO SLEEP. WHEN THIS CLOCK IS WORKING PROPERLY, YOU FEEL ALERT DURING THE DAY AND SLEEPY AT NIGHT. WHEN IT BREAKS, EVERYTHING FALLS APART. CIRCADIAN RHYTHM DISORDERS OCCUR WHEN YOUR INTERNAL BODY CLOCK IS MISALIGNED WITH THE EXTERNAL ENVIRONMENT OR WITH YOUR DESIRED SLEEP SCHEDULE. THEY ARE MORE COMMON THAN MOST PEOPLE REALIZE AND CAN BE PROFOUNDLY DISRUPTIVE.

HOW YOUR BODY CLOCK WORKS

THE CIRCADIAN CLOCK RUNS ON A ROUGHLY TWENTY-FOUR-HOUR CYCLE. IT IS SYNCHRONIZED TO THE EXTERNAL WORLD PRIMARILY THROUGH LIGHT. WHEN LIGHT ENTERS YOUR EYES, IT SIGNALS TO THE SUPRACHIASMATIC NUCLEUS THAT IT IS DAYTIME. THE SUPRACHIASMATIC NUCLEUS THEN SUPPRESSES MELATONIN PRODUCTION, RAISING YOUR ALERTNESS. AS DARKNESS FALLS, THE SUPRACHIASMATIC NUCLEUS ALLOWS MELATONIN TO RISE, PREPARING YOUR BODY FOR SLEEP. THIS SYSTEM IS REMARKABLY CONSISTENT. MOST PEOPLE'S CIRCADIAN CLOCKS RUN SLIGHTLY LONGER THAN TWENTY-FOUR HOURS, SO DAILY EXPOSURE TO MORNING LIGHT IS ESSENTIAL FOR KEEPING THEM SYNCHRONIZED. WITHOUT THAT LIGHT CUE, THE CLOCK WOULD DRIFT LATER AND

LATER, CAUSING A PROGRESSIVE DELAY IN SLEEP TIMING. DELAYED SLEEP-WAKE PHASE DISORDER DELAYED SLEEP-WAKE PHASE DISORDER, OR DSPD, IS THE MOST COMMON CIRCADIAN RHYTHM DISORDER. PEOPLE WITH DSPD HAVE A BODY CLOCK THAT IS SHIFTED SIGNIFICANTLY LATER THAN THE TYPICAL SCHEDULE. THEY NATURALLY FEEL MOST ALERT IN THE LATE EVENING AND HAVE DIFFICULTY FALLING ASLEEP UNTIL THE EARLY MORNING HOURS, TYPICALLY BETWEEN 1 A.M. AND 4 A.M. WHEN LEFT TO THEIR OWN SCHEDULE, THEY SLEEP NORMALLY BUT ON A DELAYED TIMETABLE. THE PROBLEM ARISES WHEN THEY NEED TO CONFORM TO A TYPICAL SCHEDULE. WAKING UP FOR WORK OR SCHOOL AT 7 A.M. AFTER FALLING ASLEEP AT 3 A.M. CREATES CHRONIC SLEEP DEPRIVATION. PEOPLE WITH DSPD ARE OFTEN LABELED LAZY OR UNDISCIPLINED, BUT THE REALITY IS THAT THEIR BIOLOGY IS FIGHTING AGAINST THE CLOCK. DSPD IS MOST COMMON IN ADOLESCENTS AND YOUNG ADULTS, WITH AN ESTIMATED PREVALENCE OF 7 TO 16 PERCENT IN THIS AGE GROUP. IT OFTEN PERSISTS INTO ADULTHOOD, THOUGH IT CAN IMPROVE WITH AGE. THERE IS A STRONG GENETIC COMPONENT — IF YOU HAVE DSPD, YOUR FAMILY MEMBERS ARE MORE LIKELY TO HAVE IT TOO. ADVANCED SLEEP-WAKE PHASE DISORDER ADVANCED SLEEP-WAKE PHASE DISORDER, OR ASPD, IS THE OPPOSITE OF DSPD. PEOPLE WITH ASPD HAVE A BODY CLOCK THAT IS SHIFTED EARLIER. THEY NATURALLY FEEL SLEEPY IN THE EARLY EVENING, TYPICALLY BETWEEN 6 P.M. AND 9 P.M., AND WAKE UP VERY EARLY, BETWEEN 2 A.M. AND 5 A.M. WHILE THIS MIGHT SOUND IDEAL FOR EARLY RISERS, IT CAN BE SOCIALLY ISOLATING AND DISRUPTIVE. PEOPLE WITH ASPD OFTEN STRUGGLE TO STAY AWAKE FOR EVENING SOCIAL EVENTS AND MAY FIND THEMSELVES WAKING UP LONG BEFORE THE REST OF THE WORLD. ASPD IS MORE COMMON IN OLDER ADULTS, AFFECTING AN ESTIMATED 1 PERCENT OF PEOPLE

OVER SIXTY-FIVE. IT IS LESS COMMON THAN DSPD AND MAY HAVE A GENETIC COMPONENT AS WELL. SHIFT WORK SLEEP DISORDER

SHIFT WORK SLEEP DISORDER AFFECTS PEOPLE WHO WORK DURING THE NIGHT OR ROTATE BETWEEN DAY AND NIGHT SHIFTS. THEIR WORK SCHEDULE REQUIRES THEM TO BE AWAKE WHEN THEIR BODY CLOCK SAYS IT IS TIME TO SLEEP AND ASLEEP WHEN THEIR BODY CLOCK SAYS IT IS TIME TO BE AWAKE. THE RESULT IS CHRONIC MISALIGNMENT BETWEEN THEIR INTERNAL CLOCK AND THEIR EXTERNAL DEMANDS. SHIFT WORKERS FACE A UNIQUE CHALLENGE. EVEN WHEN THEY TRY TO SLEEP DURING THE DAY, THEIR CIRCADIAN CLOCK IS PUMPING OUT ALERTING SIGNALS. DAYTIME SLEEP IS TYPICALLY SHORTER, LIGHTER, AND LESS RESTORATIVE THAN NIGHTTIME SLEEP. OVER TIME, SHIFT WORKERS ACCUMULATE A SIGNIFICANT SLEEP DEBT. THE HEALTH CONSEQUENCES OF SHIFT WORK ARE WELL DOCUMENTED. SHIFT WORKERS HAVE HIGHER RATES OF CARDIOVASCULAR DISEASE, METABOLIC DISORDERS, GASTROINTESTINAL PROBLEMS, DEPRESSION, AND CERTAIN TYPES OF CANCER. THE WORLD HEALTH ORGANIZATION HAS CLASSIFIED NIGHT SHIFT WORK AS A PROBABLE CARCINOGEN. THE RISK IS REAL AND SERIOUS.

JET LAG DISORDER

JET LAG IS A TEMPORARY CIRCADIAN RHYTHM DISORDER CAUSED BY RAPID TRAVEL ACROSS MULTIPLE TIME ZONES. WHEN YOU FLY FROM NEW YORK TO LONDON, YOUR BODY CLOCK IS STILL ON NEW YORK TIME, BUT THE ENVIRONMENT SAYS IT IS MORNING. YOUR BODY WANTS TO SLEEP, BUT THE SUN SAYS WAKE UP. THE RESULT IS FATIGUE, INSOMNIA, DIFFICULTY CONCENTRATING, DIGESTIVE PROBLEMS, AND A GENERAL FEELING OF BEING OUT OF SYNC. JET LAG TYPICALLY RESOLVES ON ITS OWN AT A RATE OF ABOUT ONE DAY PER TIME ZONE CROSSED. EASTWARD TRAVEL IS GENERALLY HARDER THAN WESTWARD TRAVEL BECAUSE IT REQUIRES ADVANCING YOUR CLOCK

— A MORE DIFFICULT ADJUSTMENT FOR MOST PEOPLE. NON-24-HOUR SLEEP-WAKE DISORDER NON-24-HOUR SLEEP-WAKE DISORDER IS A RARE CIRCADIAN RHYTHM DISORDER IN WHICH THE BODY CLOCK RUNS ON A CYCLE LONGER THAN TWENTY-FOUR HOURS. EACH DAY, THE PERSON'S NATURAL SLEEP TIME SHIFTS LATER AND LATER, CYCLING THROUGH ALL POSSIBLE SLEEP TIMES OVER THE COURSE OF WEEKS OR MONTHS. THIS DISORDER IS MOST COMMON IN PEOPLE WHO ARE TOTALLY BLIND. WITHOUT LIGHT PERCEPTION, THE CIRCADIAN CLOCK CANNOT BE SYNCHRONIZED TO THE TWENTY-FOUR-HOUR DAY. IT AFFECTS ABOUT 50 TO 75 PERCENT OF BLIND INDIVIDUALS. IT CAN ALSO OCCUR IN SIGHTED PEOPLE, THOUGH THIS IS RARE. TREATING CIRCADIAN RHYTHM DISORDERS THE PRIMARY TREATMENT FOR CIRCADIAN RHYTHM DISORDERS IS TIMED LIGHT EXPOSURE. LIGHT IS THE MOST POWERFUL SYNCHRONIZER OF THE CIRCADIAN CLOCK, AND CAREFUL TIMING OF LIGHT EXPOSURE CAN SHIFT THE CLOCK IN THE DESIRED DIRECTION. FOR DSPD, MORNING BRIGHT LIGHT THERAPY IS THE CORNERSTONE OF TREATMENT. EXPOSURE TO BRIGHT LIGHT SHORTLY AFTER WAKING HELPS ADVANCE THE CLOCK, MAKING IT EASIER TO FALL ASLEEP EARLIER IN THE EVENING. LIGHT BOXES THAT DELIVER 10,000 LUX OF COOL WHITE LIGHT ARE COMMONLY USED. TREATMENT TYPICALLY INVOLVES THIRTY TO SIXTY MINUTES OF LIGHT EXPOSURE IMMEDIATELY UPON WAKING. MELATONIN SUPPLEMENTATION CAN ALSO HELP. A SMALL DOSE OF MELATONIN, TYPICALLY 0.5 TO 3 MILLIGRAMS, TAKEN A FEW HOURS BEFORE THE DESIRED BEDTIME, CAN HELP ADVANCE THE CLOCK. THE TIMING IS CRITICAL — MELATONIN TAKEN TOO EARLY OR TOO LATE CAN HAVE THE OPPOSITE EFFECT. FOR SHIFT WORKERS, STRATEGIC USE OF LIGHT AND DARK IS ESSENTIAL. WEARING BLUE-BLOCKING GLASSES DURING THE COMMUTE HOME FROM A NIGHT SHIFT CAN PREVENT LIGHT FROM

SIGNALING WAKEFULNESS. CREATING A COMPLETELY DARK, QUIET, AND COOL SLEEPING ENVIRONMENT DURING THE DAY IS CRUCIAL. STRATEGIC NAPPING BEFORE A NIGHT SHIFT CAN IMPROVE ALERTNESS DURING WORK. PRACTICAL STRATEGIES IF YOU SUSPECT YOU HAVE A CIRCADIAN RHYTHM DISORDER, START BY TRACKING YOUR NATURAL SLEEP PATTERNS FOR TWO WEEKS. NOTE WHEN YOU NATURALLY FALL ASLEEP AND WAKE UP ON DAYS WHEN YOU HAVE NO SCHEDULE CONSTRAINTS. THIS WILL REVEAL YOUR NATURAL CHRONOTYPE. THEN, WORK WITH A SLEEP SPECIALIST TO DEVELOP A TREATMENT PLAN. CIRCADIAN RHYTHM DISORDERS ARE HIGHLY TREATABLE, BUT THE TREATMENT REQUIRES PRECISION — THE TIMING OF LIGHT EXPOSURE AND MELATONIN MATTERS ENORMOUSLY. REFLECTION QUESTIONS

What is your natural sleep schedule when you have no obligations the next day? Do you find it easier to stay up late or wake up early? Have you ever worked night shifts or rotating shifts? How did it affect your sleep and health? What is the biggest challenge you face in maintaining a consistent sleep schedule?

Social Jet Lag Social jet lag is a term that describes the misalignment between your internal biological clock and the social schedule imposed by work, school, and social obligations. It is most common in people who sleep later on weekends than on weekdays, then struggle to wake up on Monday morning. The effects of social jet lag go beyond Monday morning grogginess. Research has found that even one hour of social jet lag — a one-hour difference between your natural sleep

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CHAPTER 7: THE ANXIETY-SLEEP LOOP

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ANXIETY AND SLEEP HAVE A COMPLICATED RELATIONSHIP. THEY ARE DEEPLY INTERCONNECTED, EACH INFLUENCING THE OTHER IN WAYS THAT CAN CREATE A SELF-REINFORCING CYCLE. ANXIETY MAKES IT HARD TO SLEEP. POOR SLEEP MAKES ANXIETY WORSE. AND ONCE THIS LOOP GETS STARTED, IT CAN BE REMARKABLY DIFFICULT TO BREAK. HOW ANXIETY DISRUPTS SLEEP

ANXIETY IS A STATE OF HEIGHTENED ALERTNESS. IT IS YOUR BODY'S THREAT DETECTION SYSTEM GOING INTO OVERDRIVE. WHEN YOU ARE ANXIOUS, YOUR SYMPATHETIC NERVOUS SYSTEM IS ACTIVATED, RELEASING STRESS HORMONES LIKE CORTISOL AND ADRENALINE. YOUR HEART RATE INCREASES, YOUR BREATHING BECOMES SHALLOW, AND YOUR MUSCLES TENSE. YOU ARE PRIMED FOR ACTION. THIS IS THE EXACT OPPOSITE OF THE STATE YOU NEED TO FALL ASLEEP. SLEEP REQUIRES PARASYMPATHETIC ACTIVATION — THE REST-AND-DIGEST SYSTEM. IT REQUIRES RELAXATION, LOWERED HEART RATE, AND A SENSE OF SAFETY. WHEN YOUR NERVOUS SYSTEM IS IN FIGHT-OR-FLIGHT MODE, SLEEP BECOMES NEARLY IMPOSSIBLE. ANXIETY KEEPS THE BRAIN IN A STATE OF HYPERVIGILANCE. IT SCANS FOR THREATS, REPLAYS WORRYING SCENARIOS, AND CREATES CATASTROPHIC PREDICTIONS ABOUT THE FUTURE. IN THE QUIET OF THE BEDROOM, WITH NO EXTERNAL DISTRACTIONS, THESE THOUGHTS BECOME AMPLIFIED. THE MIND SPINS, AND SLEEP RECEDES FURTHER INTO THE DISTANCE. HOW POOR SLEEP FUELS ANXIETY

THE RELATIONSHIP WORKS IN THE OTHER DIRECTION TOO. POOR SLEEP

INCREASES ANXIETY. RESEARCH HAS FOUND THAT EVEN ONE NIGHT OF SLEEP DEPRIVATION CAN INCREASE ANXIETY LEVELS BY 30 PERCENT. BRAIN IMAGING STUDIES SHOW THAT SLEEP DEPRIVATION AMPLIFIES ACTIVITY IN THE AMYGDALA, THE BRAIN'S FEAR CENTER, WHILE REDUCING ACTIVITY IN THE PREFRONTAL CORTEX, WHICH REGULATES EMOTIONAL RESPONSES. WHEN YOU ARE SLEEP-DEPRIVED, YOUR EMOTIONAL REGULATION SUFFERS. SMALL STRESSORS FEEL OVERWHELMING. MINOR FRUSTRATIONS TRIGGER DISPROPORTIONATE REACTIONS. THE FILTER THAT NORMALLY HELPS YOU DISTINGUISH BETWEEN GENUINE THREATS AND MINOR ANNOYANCES BECOMES IMPAIRED. EVERYTHING FEELS MORE THREATENING THAN IT ACTUALLY IS. THIS CREATES A DANGEROUS FEEDBACK LOOP. YOU CANNOT SLEEP BECAUSE YOU ARE ANXIOUS. YOU ARE MORE ANXIOUS BECAUSE YOU CANNOT SLEEP. EACH REINFORCES THE OTHER, AND THE LOOP TIGHTENS WITH EVERY CYCLE. THE CATASTROPHIZING SPIRAL ONE OF THE MOST COMMON MANIFESTATIONS OF THE ANXIETY-SLEEP LOOP IS CATASTROPHIZING ABOUT SLEEP ITSELF. YOU LIE IN BED, AND AS THE MINUTES TICK BY, YOUR THOUGHTS SPIRAL. "IF I DO NOT FALL ASLEEP SOON, I WILL ONLY GET FIVE HOURS. THEN I WILL BE EXHAUSTED AT WORK TOMORROW. I HAVE AN IMPORTANT MEETING. I WILL PERFORM TERRIBLY. I MIGHT LOSE MY JOB. MY LIFE IS FALLING APART BECAUSE I CANNOT SLEEP." THIS CATASTROPHIZING IS UNDERSTANDABLE, BUT IT MAKES THE SITUATION WORSE. THE ANXIETY ABOUT NOT SLEEPING CREATES MORE HYPERAROUSAL, WHICH MAKES SLEEP EVEN HARDER TO ACHIEVE. YOU ARE NOW FIGHTING A BATTLE ON TWO FRONTS — THE ORIGINAL ANXIETY THAT STARTED THE PROBLEM AND THE ANXIETY ABOUT THE ANXIETY. GENERALIZED ANXIETY AND SLEEP PEOPLE WITH GENERALIZED

ANXIETY DISORDER, OR GAD, ARE AT PARTICULARLY HIGH RISK FOR SLEEP PROBLEMS. GAD IS CHARACTERIZED BY EXCESSIVE, UNCONTROLLABLE WORRY ABOUT A WIDE RANGE OF TOPICS. THIS WORRY DOES NOT STOP AT BEDTIME. IN FACT, IT OFTEN INTENSIFIES BECAUSE THERE ARE FEWER DISTRACTIONS. RESEARCH HAS FOUND THAT PEOPLE WITH GAD HAVE HIGHER PHYSIOLOGICAL AROUSAL DURING SLEEP — THEIR HEART RATE AND CORTISOL LEVELS REMAIN ELEVATED EVEN WHEN THEY ARE ASLEEP. THEY SPEND LESS TIME IN DEEP SLEEP AND MORE TIME IN LIGHTER STAGES. THEIR SLEEP IS LESS RESTORATIVE, EVEN WHEN THEY GET ADEQUATE DURATION. PANIC DISORDER AND SLEEP PANIC DISORDER CAN ALSO DISRUPT SLEEP IN DISTINCTIVE WAYS. NOCTURNAL PANIC ATTACKS OCCUR DURING SLEEP, TYPICALLY DURING THE TRANSITION FROM STAGE TWO TO DEEP SLEEP. THE PERSON WAKES UP IN A STATE OF INTENSE FEAR, WITH A RACING HEART, SHORTNESS OF BREATH, CHEST PAIN, AND A SENSE OF IMPENDING DOOM. THESE ATTACKS CAN BE TERRIFYING AND CAN CREATE A LASTING FEAR OF FALLING ASLEEP. UNLIKE NIGHTMARES, NOCTURNAL PANIC ATTACKS OCCUR WITHOUT PRECEDING DREAM CONTENT. THE PERSON WAKES UP ALREADY IN A STATE OF PANIC, OFTEN WITH NO MEMORY OF WHAT TRIGGERED IT. THE FEAR OF HAVING ANOTHER ATTACK CAN CREATE SIGNIFICANT SLEEP AVOIDANCE. THE ROLE OF RACING THOUGHTS RACING THOUGHTS AT BEDTIME ARE ONE OF THE MOST COMMON COMPLAINTS LINKING ANXIETY AND INSOMNIA. THESE THOUGHTS ARE OFTEN REPETITIVE AND WORRY-BASED. THEY REPLAY PAST EVENTS, ANTICIPATE FUTURE PROBLEMS, AND GENERATE WORST-CASE SCENARIOS. THEY FEEL INVOLUNTARY AND UNCONTROLLABLE. THE CONTENT OF THESE THOUGHTS MATTERS LESS THAN THE PATTERN. THE BRAIN IS IN A LOOP. IT KEEPS CIRCLING BACK TO THE

SAME CONCERNS, PROCESSING AND REPROCESSING WITHOUT RESOLUTION. THE LACK OF RESOLUTION KEEPS THE ANXIETY ALIVE, WHICH KEEPS THE AROUSAL HIGH, WHICH PREVENTS SLEEP.

BREAKING THE LOOP BREAKING THE ANXIETY-SLEEP LOOP REQUIRES ADDRESSING BOTH SIDES OF THE EQUATION. MANAGING ANXIETY REDUCES ITS IMPACT ON SLEEP. IMPROVING SLEEP REDUCES ITS CONTRIBUTION TO ANXIETY. THE MOST EFFECTIVE APPROACHES WORK ON BOTH FRONTS SIMULTANEOUSLY. COGNITIVE BEHAVIORAL THERAPY IS EFFECTIVE FOR BOTH ANXIETY AND INSOMNIA. IT HELPS YOU IDENTIFY AND CHALLENGE THE THOUGHT PATTERNS THAT FUEL ANXIETY AND THE BEHAVIORS THAT PERPETUATE INSOMNIA. THE SKILLS YOU LEARN FOR MANAGING ANXIOUS THOUGHTS CAN BE APPLIED DIRECTLY TO THE RACING THOUGHTS THAT KEEP YOU AWAKE AT NIGHT. RELAXATION TECHNIQUES CAN HELP CALM THE NERVOUS SYSTEM BEFORE BED. DEEP BREATHING EXERCISES ACTIVATE THE PARASYMPATHETIC NERVOUS SYSTEM. PROGRESSIVE MUSCLE RELAXATION REDUCES PHYSICAL TENSION. GUIDED IMAGERY PROVIDES A MENTAL FOCUS THAT DISPLACES ANXIOUS THOUGHTS. THESE TECHNIQUES ARE NOT CURES, BUT THEY ARE TOOLS THAT CAN HELP YOU SHIFT YOUR NERVOUS SYSTEM TOWARD A STATE MORE CONDUCTIVE TO SLEEP. ESTABLISHING A CONSISTENT WIND-DOWN ROUTINE IS PARTICULARLY IMPORTANT FOR PEOPLE WITH ANXIETY. THE TRANSITION FROM DAYTIME ALERTNESS TO NIGHTTIME RESTFULNESS SHOULD BE GRADUAL. GIVE YOURSELF AT LEAST AN HOUR OF WIND-DOWN TIME, FREE FROM SCREENS, WORK, AND STIMULATING ACTIVITIES. USE THIS TIME FOR CALMING ACTIVITIES THAT SIGNAL SAFETY TO YOUR NERVOUS SYSTEM.

THE WORRY TIME TECHNIQUE ONE SPECIFIC TECHNIQUE THAT CAN HELP WITH BEDTIME RACING THOUGHTS IS SCHEDULED WORRY TIME. SET ASIDE FIFTEEN TO THIRTY MINUTES EARLIER IN THE DAY, AT THE SAME

TIME EVERY DAY, TO ACTIVELY WORRY. WRITE DOWN YOUR CONCERNS, THINK THROUGH THEM, AND IF POSSIBLE, IDENTIFY ACTION STEPS. WHEN A WORRYING THOUGHT ARISES AT BEDTIME, REMIND YOURSELF THAT YOU HAVE ALREADY ADDRESSED IT DURING YOUR WORRY TIME AND CAN RETURN TO IT TOMORROW. THIS TECHNIQUE IS NOT ABOUT SUPPRESSING WORRY. IT IS ABOUT CONTAINING IT. BY GIVING YOUR WORRY A DESIGNATED TIME AND PLACE, YOU PREVENT IT FROM INVADING YOUR SLEEP TIME WHILE STILL ALLOWING YOURSELF TO PROCESS LEGITIMATE CONCERNS.

WHEN TO SEEK PROFESSIONAL HELP IF ANXIETY IS SIGNIFICANTLY DISRUPTING YOUR SLEEP ON A REGULAR BASIS, IT IS WORTH SEEKING PROFESSIONAL SUPPORT. THERAPY, PARTICULARLY COGNITIVE BEHAVIORAL THERAPY, CAN BE HIGHLY EFFECTIVE FOR BOTH ANXIETY AND INSOMNIA. IN SOME CASES, MEDICATION MAY BE APPROPRIATE, EITHER FOR THE ANXIETY OR FOR THE SLEEP DISRUPTION IT CAUSES. THE ANXIETY-SLEEP LOOP IS ONE OF THE MOST COMMON SLEEP PROBLEMS, AND IT IS ALSO ONE OF THE MOST TREATABLE. WITH THE RIGHT APPROACH, YOU CAN BREAK THE CYCLE.

REFLECTION QUESTIONS

When you lie in bed at night, what kinds of thoughts tend to keep you awake? Have you noticed that your anxiety is worse after a night of poor sleep? What helps you feel calmest and safest at the end of the day? Have you ever tried any relaxation techniques before bed? What was your experience?

The Physiological Connection The anxiety-sleep connection is not just psychological — it is deeply physiological. Anxiety activates the hypothalamic-pituitary-adrenal axis, the body's central stress response system. This activation leads to increased cortisol production, which follows a

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CHAPTER 8: PARASOMNIAS — WHEN SLEEP GETS STRANGE

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PARASOMNIAS ARE A CATEGORY OF SLEEP DISORDERS THAT INVOLVE UNUSUAL BEHAVIORS, MOVEMENTS, EMOTIONS, PERCEPTIONS, OR DREAMS THAT OCCUR WHILE FALLING ASLEEP, SLEEPING, OR WAKING UP. THEY ARE ESSENTIALLY GLITCHES IN THE BOUNDARIES BETWEEN SLEEP STATES — MOMENTS WHEN THE BRAIN IS PARTIALLY IN ONE STATE AND PARTIALLY IN ANOTHER, PRODUCING BEHAVIORS THAT THE CONSCIOUS MIND WOULD NEVER ENDORSE. PARASOMNIAS ARE MORE COMMON IN CHILDREN, BUT THEY AFFECT ADULTS TOO. THEY CAN RANGE FROM MILDLY AMUSING TO GENUINELY DANGEROUS, AND THEY OFTEN HAVE A SIGNIFICANT IMPACT ON BOTH THE PERSON EXPERIENCING THEM AND THEIR BED PARTNER.

SLEEPWALKING

SLEEPWALKING, OR SOMNAMBULISM, IS ONE OF THE MOST WELL-KNOWN PARASOMNIAS. IT INVOLVES GETTING UP AND WALKING AROUND WHILE IN A STATE OF DEEP SLEEP. THE PERSON'S EYES ARE OPEN, BUT THEY ARE NOT CONSCIOUS. THEY MAY PERFORM COMPLEX BEHAVIORS — DRESSING, EATING, MOVING FURNITURE, EVEN LEAVING THE HOUSE — WITHOUT ANY AWARENESS. SLEEPWALKING OCCURS DURING DEEP SLEEP, TYPICALLY IN THE FIRST THIRD OF THE NIGHT. THE BRAIN IS IN A MIXED STATE — THE PARTS THAT CONTROL MOVEMENT ARE ACTIVE, BUT THE PARTS THAT CONTROL CONSCIOUSNESS, MEMORY, AND JUDGMENT ARE NOT. THIS IS WHY SLEEPWALKERS CAN NAVIGATE THEIR ENVIRONMENT BUT HAVE NO MEMORY OF THEIR ACTIONS THE NEXT DAY. CONTRARY TO POPULAR BELIEF, IT IS NOT

DANGEROUS TO WAKE A SLEEPWALKER. THE RISK IS THAT WAKING THEM ABRUPTLY CAN CAUSE CONFUSION AND DISTRESS, POTENTIALLY LEADING TO AGGRESSIVE BEHAVIOR. THE BETTER APPROACH IS TO GUIDE THEM GENTLY BACK TO BED. SLEEPWALKING IS MORE COMMON IN CHILDREN, AFFECTING UP TO 17 PERCENT OF KIDS, AND USUALLY RESOLVES ON ITS OWN. IN ADULTS, IT AFFECTS ABOUT 2 TO 4 PERCENT AND CAN BE TRIGGERED BY SLEEP DEPRIVATION, STRESS, FEVER, ALCOHOL, AND CERTAIN MEDICATIONS. SLEEP TALKING SLEEP TALKING, OR SOMNILOQUY, IS ONE OF THE MOST COMMON AND LEAST CONCERNING PARASOMNIAS. ALMOST EVERYONE TALKS IN THEIR SLEEP AT SOME POINT. THE SPEECH CAN RANGE FROM MUMBLES AND GIBBERISH TO FULL, COHERENT SENTENCES. THE CONTENT IS OFTEN MUNDANE, THOUGH IT CAN BE EMOTIONAL OR PROFANE. SLEEP TALKING CAN OCCUR IN ANY SLEEP STAGE, BUT THE TYPE OF SPEECH VARIES. IN NREM SLEEP, TALKING TENDS TO BE SIMPLER AND LESS STORY-LIKE. IN REM SLEEP, IT CAN BE MORE COMPLEX AND MAY BE RELATED TO DREAM CONTENT. SLEEP TALKING RARELY REQUIRES TREATMENT. IT IS USUALLY HARMLESS, THOUGH IT CAN BE EMBARRASSING OR DISRUPTIVE FOR BED PARTNERS. REDUCING STRESS, MAINTAINING A CONSISTENT SLEEP SCHEDULE, AND AVOIDING ALCOHOL BEFORE BED CAN REDUCE ITS FREQUENCY. NIGHT TERRORS NIGHT TERRORS ARE DISTINCT FROM NIGHTMARES. A NIGHTMARE IS A BAD DREAM THAT OCCURS DURING REM SLEEP, AND THE PERSON WAKES UP REMEMBERING IT. A NIGHT TERROR OCCURS DURING DEEP SLEEP, AND THE PERSON REMAINS ASLEEP THROUGHOUT THE EPISODE. THEY MAY SIT UP IN BED, SCREAM, THRASH, AND APPEAR TERRIFIED, BUT THEY ARE NOT CONSCIOUS AND WILL HAVE NO MEMORY OF THE EPISODE THE NEXT MORNING.

NIGHT TERRORS ARE MOST COMMON IN CHILDREN BETWEEN THE AGES OF THREE AND SEVEN, AFFECTING ABOUT 3 TO 6 PERCENT OF KIDS. IN ADULTS, THEY ARE LESS COMMON AND OFTEN LINKED TO STRESS, TRAUMA, SLEEP DEPRIVATION, OR ALCOHOL USE. ADULT NIGHT TERRORS CAN BE MORE INTENSE AND MAY INVOLVE AGGRESSIVE OR VIOLENT BEHAVIORS. LIKE SLEEPWALKING, NIGHT TERRORS ARE A DISORDER OF AROUSAL — THE BRAIN GETS STUCK BETWEEN DEEP SLEEP AND WAKEFULNESS. THE PERSON APPEARS AWAKE BUT IS NOT. ATTEMPTING TO WAKE THEM DURING A NIGHT TERROR IS GENERALLY NOT HELPFUL AND CAN PROLONG THE EPISODE.

REM SLEEP BEHAVIOR DISORDER REM SLEEP BEHAVIOR DISORDER, OR RBD, IS A PARASOMNIA IN WHICH THE NORMAL PARALYSIS THAT OCCURS DURING REM SLEEP IS ABSENT. THIS MEANS THE PERSON PHYSICALLY ACTS OUT THEIR DREAMS, OFTEN WITH DRAMATIC AND SOMETIMES VIOLENT MOVEMENTS. DURING REM SLEEP, THE BRAIN SENDS A SIGNAL TO THE SPINAL CORD THAT TEMPORARILY PARALYZES MOST VOLUNTARY MUSCLES. THIS PARALYSIS, CALLED REM ATONIA, PREVENTS YOU FROM ACTING OUT YOUR DREAMS. IN RBD, THIS SYSTEM FAILS. THE PERSON MAY PUNCH, KICK, LEAP OUT OF BED, OR PERFORM COMPLEX MOVEMENTS THAT CORRESPOND TO THE CONTENT OF THEIR DREAMS. RBD IS DIFFERENT FROM SLEEPWALKING. IN SLEEPWALKING, THE PERSON IS IN DEEP SLEEP AND HAS NO CONSCIOUS AWARENESS. IN RBD, THE PERSON IS IN REM SLEEP AND IS ACTING OUT A DREAM. THEY CAN OFTEN RECALL THE DREAM CONTENT WHEN WOKEN UP. RBD IS STRONGLY ASSOCIATED WITH NEURODEGENERATIVE DISORDERS, PARTICULARLY PARKINSON'S DISEASE AND LEWY BODY DEMENTIA. IN FACT, RBD CAN PRECEDE THE ONSET OF THESE CONDITIONS BY YEARS OR EVEN DECADES. ANYONE WHO DEVELOPS RBD SHOULD UNDERGO A NEUROLOGICAL

EVALUATION. TREATMENT WITH CLONAZEPAM OR MELATONIN IS OFTEN EFFECTIVE. SLEEP EATING DISORDER SLEEP EATING DISORDER, OR NS-RED, IS A PARASOMNIA THAT INVOLVES EATING WHILE IN A STATE OF PARTIAL AROUSAL FROM SLEEP. THE PERSON MAY PREPARE AND CONSUME FOOD, OFTEN STRANGE COMBINATIONS THAT THEY WOULD NOT EAT WHILE AWAKE, WITH NO MEMORY OF IT IN THE MORNING. THIS CAN LEAD TO WEIGHT GAIN, HEALTH PROBLEMS FROM CONSUMING INAPPROPRIATE FOODS, AND SAFETY RISKS FROM COOKING WHILE NOT FULLY CONSCIOUS. SLEEP EATING DISORDER IS MORE COMMON IN WOMEN AND IS OFTEN ASSOCIATED WITH OTHER SLEEP DISORDERS, PARTICULARLY SLEEPWALKING AND RESTLESS LEGS SYNDROME. IT CAN ALSO BE TRIGGERED BY CERTAIN SLEEP MEDICATIONS, PARTICULARLY ZOLPIDEM. CONFUSIONAL AROUSALS CONFUSIONAL AROUSALS ARE EPISODES OF CONFUSION AND DISORIENTATION UPON WAKING. THE PERSON MAY APPEAR AWAKE BUT IS NOT FULLY CONSCIOUS. THEY MAY SPEAK INCOHERENTLY, MOVE SLOWLY, AND SEEM CONFUSED ABOUT WHERE THEY ARE OR WHAT IS HAPPENING. THESE EPISODES ARE MOST COMMON IN CHILDREN AND TYPICALLY LAST A FEW MINUTES. IN ADULTS, CONFUSIONAL AROUSALS CAN BE TRIGGERED BY SLEEP DEPRIVATION, ALCOHOL, AND SLEEPING IN AN UNFAMILIAR ENVIRONMENT. THEY ARE GENERALLY HARMLESS BUT CAN BE DISTRESSING FOR THE PERSON EXPERIENCING THEM. MANAGING PARASOMNIAS THE MANAGEMENT OF PARASOMNIAS DEPENDS ON THE TYPE AND SEVERITY. FOR MOST PARASOMNIAS IN CHILDREN, REASSURANCE AND SAFETY MEASURES ARE SUFFICIENT. MAKING THE SLEEP ENVIRONMENT SAFE — LOCKING DOORS AND WINDOWS, REMOVING SHARP OBJECTS, INSTALLING GATES AT THE TOP OF STAIRS — CAN PREVENT INJURIES. IN ADULTS, IDENTIFYING AND

ADDRESSING TRIGGERS IS THE FIRST STEP. SLEEP DEPRIVATION IS A COMMON TRIGGER ACROSS MOST PARASOMNIAS. ADDRESSING THE UNDERLYING SLEEP QUALITY ISSUES CAN REDUCE PARASOMNIA FREQUENCY. ALCOHOL AVOIDANCE, STRESS MANAGEMENT, AND A CONSISTENT SLEEP SCHEDULE ARE IMPORTANT. FOR DANGEROUS PARASOMNIAS, PARTICULARLY RBD AND VIOLENT SLEEPWALKING, MEDICATION MAY BE NECESSARY. CLONAZEPAM IS THE MOST COMMONLY PRESCRIBED MEDICATION FOR RBD AND HAS SHOWN GOOD EFFICACY. MELATONIN IS AN ALTERNATIVE WITH FEWER SIDE EFFECTS. WHEN TO SEE A SPECIALIST MOST PARASOMNIAS ARE HARMLESS AND DO NOT REQUIRE TREATMENT. HOWEVER, YOU SHOULD SEE A SLEEP SPECIALIST IF THE BEHAVIORS ARE DANGEROUS, IF THEY ARE CAUSING SIGNIFICANT DISRUPTION TO YOUR OR YOUR PARTNER'S SLEEP, IF THEY HAVE RESULTED IN INJURY, OR IF THEY STARTED SUDDENLY IN ADULthood. A SLEEP STUDY CAN HELP DIFFERENTIATE BETWEEN DIFFERENT TYPES OF PARASOMNIAS AND RULE OUT OTHER CONDITIONS LIKE SLEEP APNEA OR SEIZURES. REFLECTION QUESTIONS

Have you ever been told that you do unusual things in your sleep? What did your bed partner describe? Do you remember your dreams? Have you ever woken up with injuries that you could not explain? Are there patterns to your parasomnias — do they happen more often when you are stressed or sleep-deprived? Have you taken any safety measures to protect yourself during sleep?

Sleep-Related Eating Disorder in Depth Sleep-related eating disorder, or SRED, is a fascinating and often distressing parasomnia. People with SRED eat while in a state of mixed consciousness, typically during partial arousal

from deep sleep. They may consume unusual combinations of food, sometimes raw ingredients or even non-food items, and have little t

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CHAPTER 9: THE SCIENCE OF SLEEP HYGIENE THAT ACTUALLY WORKS

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SLEEP HYGIENE IS THE TERM USED TO DESCRIBE THE HABITS AND ENVIRONMENTAL FACTORS THAT PROMOTE GOOD SLEEP. IT IS A SIMPLE CONCEPT, BUT IT IS OFTEN PRESENTED IN A SIMPLISTIC WAY. TELLING SOMEONE WITH CHRONIC INSOMNIA TO "AVOID CAFFEINE BEFORE BED" IS LIKE TELLING SOMEONE WITH DEPRESSION TO "THINK POSITIVE THOUGHTS." IT IS NOT WRONG, BUT IT IS RARELY ENOUGH ON ITS OWN. THIS CHAPTER GOES BEYOND THE BASICS. IT EXPLAINS NOT JUST WHAT THE SCIENCE SAYS ABOUT SLEEP HYGIENE, BUT WHY EACH RECOMMENDATION WORKS AND HOW TO IMPLEMENT IT EFFECTIVELY.

THE FOUNDATION: CONSISTENCY

THE SINGLE MOST IMPORTANT ELEMENT OF SLEEP HYGIENE IS CONSISTENCY. GOING TO BED AND WAKING UP AT THE SAME TIME EVERY DAY — INCLUDING WEEKENDS — IS MORE POWERFUL THAN ANY OTHER SLEEP HABIT. CONSISTENCY TRAINS YOUR CIRCADIAN CLOCK. IT STRENGTHENS THE ASSOCIATION BETWEEN YOUR BEDTIME AND SLEEP. IT REDUCES THE VARIABILITY THAT DISRUPTS SLEEP CYCLES. MOST PEOPLE UNDERESTIMATE HOW IMPACTFUL CONSISTENCY IS. EVEN SHIFTING YOUR SLEEP SCHEDULE BY AN HOUR ON WEEKENDS CAN CREATE A PHENOMENON CALLED SOCIAL JET LAG — A MISALIGNMENT BETWEEN YOUR INTERNAL CLOCK AND YOUR SOCIAL SCHEDULE THAT LEAVES YOU FEELING TIRED ON MONDAY MORNING. THE GOAL IS NOT PERFECTION. AIM FOR CONSISTENCY WITHIN A THIRTY-MINUTE WINDOW. IF YOU USUALLY GO TO BED AT 10:30 P.M., GOING TO BED

AT 11:00 P.M. ON FRIDAY IS FINE. GOING TO BED AT 2:00 A.M. IS NOT. LIGHT EXPOSURE LIGHT IS THE MOST POWERFUL REGULATOR OF YOUR CIRCADIAN CLOCK. MANAGING YOUR LIGHT EXPOSURE IS ONE OF THE MOST EFFECTIVE THINGS YOU CAN DO FOR YOUR SLEEP. THE KEY IS GETTING BRIGHT LIGHT EARLY IN THE DAY AND DIM LIGHT IN THE EVENING. MORNING LIGHT EXPOSURE IS CRITICAL. EXPOSURE TO BRIGHT LIGHT WITHIN THE FIRST HOUR OF WAKING SIGNALS TO YOUR SUPRACHIASMATIC NUCLEUS THAT THE DAY HAS STARTED. IT SUPPRESSES MELATONIN, RAISES ALERTNESS, AND SETS YOUR CIRCADIAN CLOCK FOR THE DAY. THE IDEAL IS NATURAL SUNLIGHT, BUT A LIGHT BOX CAN WORK ON DARK MORNINGS. IN THE EVENING, THE OPPOSITE APPLIES. EXPOSURE TO BRIGHT LIGHT, PARTICULARLY BLUE LIGHT FROM SCREENS, SIGNALS TO YOUR BRAIN THAT IT IS STILL DAYTIME. THIS DELAYS MELATONIN RELEASE AND MAKES IT HARDER TO FALL ASLEEP. DIMMING LIGHTS IN THE HOUR BEFORE BED, USING BLUE-LIGHT BLOCKING GLASSES, AND AVOIDING SCREENS CAN SIGNIFICANTLY IMPROVE SLEEP ONSET. THE SCIENCE IS CLEAR: LIGHT EXPOSURE MANAGEMENT IS NOT A MINOR SUGGESTION. IT IS A FUNDAMENTAL TOOL FOR REGULATING SLEEP.

TEMPERATURE YOUR BODY TEMPERATURE PLAYS A CENTRAL ROLE IN SLEEP REGULATION. TO FALL ASLEEP, YOUR CORE BODY TEMPERATURE NEEDS TO DROP BY ABOUT ONE TO TWO DEGREES FAHRENHEIT. THIS DROP IS PART OF THE NATURAL CIRCADIAN RHYTHM. WHEN YOUR TEMPERATURE DROPS, YOU FEEL SLEEPY. WHEN IT RISES, YOU FEEL ALERT. A COOL BEDROOM SUPPORTS THIS TEMPERATURE DROP. THE OPTIMAL BEDROOM TEMPERATURE FOR SLEEP IS BETWEEN SIXTY-FIVE AND SIXTY-EIGHT DEGREES FAHRENHEIT. TEMPERATURES ABOVE SEVENTY-FIVE OR BELOW FIFTY-FOUR DEGREES CAN DISRUPT SLEEP. TAKING A WARM BATH OR SHOWER ONE TO TWO HOURS BEFORE BED CAN ACTUALLY HELP YOU

FALL ASLEEP FASTER. THE WARM WATER RAISES YOUR BODY TEMPERATURE, AND WHEN YOU GET OUT, YOUR TEMPERATURE DROPS RAPIDLY. THIS TEMPERATURE DROP MIMICS THE NATURAL COOLING THAT PRECEDES SLEEP AND CAN ACCELERATE SLEEP ONSET. NOISE AND SOUND NOISE IS ONE OF THE MOST COMMON ENVIRONMENTAL DISRUPTORS OF SLEEP. SUDDEN NOISES CAN CAUSE MICRO-AROUSALS EVEN IF THEY DO NOT FULLY WAKE YOU. CONTINUOUS NOISE CAN PREVENT YOU FROM REACHING DEEP SLEEP. WHITE NOISE, PINK NOISE, OR NATURE SOUNDS CAN HELP BY MASKING SUDDEN NOISES AND CREATING A CONSISTENT AUDITORY ENVIRONMENT. PINK NOISE, WHICH HAS MORE POWER AT LOWER FREQUENCIES THAN WHITE NOISE, HAS BEEN SHOWN TO ENHANCE DEEP SLEEP IN SOME STUDIES. IF YOU LIVE IN A NOISY ENVIRONMENT, EARPLUGS, A WHITE NOISE MACHINE, OR NOISE-CANCELING HEADPHONES CAN BE EFFECTIVE. IF YOU LIVE WITH A SNORING PARTNER, ADDRESSING THE SNORING DIRECTLY — RATHER THAN TRYING TO SLEEP THROUGH IT — IS THE BETTER LONG-TERM SOLUTION. CAFFEINE CAFFEINE IS A POWERFUL STIMULANT THAT BLOCKS ADENOSINE RECEPTORS IN THE BRAIN. ADENOSINE IS A NEUROTRANSMITTER THAT BUILDS UP DURING WAKEFULNESS AND CREATES SLEEP PRESSURE. BY BLOCKING ADENOSINE, CAFFEINE REDUCES YOUR FEELING OF SLEEPINESS. THE HALF-LIFE OF CAFFEINE IS ABOUT FIVE HOURS, MEANING THAT HALF OF THE CAFFEINE YOU CONSUME IS STILL IN YOUR SYSTEM FIVE HOURS LATER. AFTER TEN HOURS, ABOUT A QUARTER REMAINS. THIS IS WHY A CUP OF COFFEE AT 4 P.M. CAN STILL AFFECT YOUR SLEEP AT 11 P.M. THE EFFECTS OF CAFFEINE VARY SIGNIFICANTLY BETWEEN INDIVIDUALS, DUE TO GENETIC DIFFERENCES IN CAFFEINE METABOLISM. SOME PEOPLE CAN DRINK COFFEE AFTER DINNER AND SLEEP FINE. OTHERS ARE SENSITIVE ENOUGH THAT A SINGLE CUP IN

THE MORNING AFFECTS THEIR SLEEP. PAY ATTENTION TO YOUR OWN SENSITIVITY AND ADJUST ACCORDINGLY. ALCOHOL ALCOHOL IS A DOUBLE-EDGED SWORD FOR SLEEP. IT CAN MAKE YOU FALL ASLEEP FASTER BECAUSE IT IS A CENTRAL NERVOUS SYSTEM DEPRESSANT. BUT IT SIGNIFICANTLY DISRUPTS SLEEP QUALITY. ALCOHOL SUPPRESSES REM SLEEP IN THE FIRST HALF OF THE NIGHT AND CAUSES MORE FREQUENT AWAKENINGS IN THE SECOND HALF. IT ALSO RELAXES THE THROAT MUSCLES, WORSENING SNORING AND SLEEP APNEA. EVEN MODERATE ALCOHOL CONSUMPTION BEFORE BED CAN REDUCE SLEEP QUALITY. A SINGLE GLASS OF WINE MAY HELP YOU FALL ASLEEP, BUT THE SLEEP YOU GET WILL BE LESS RESTORATIVE. IF YOU CHOOSE TO DRINK, FINISHING YOUR LAST DRINK AT LEAST THREE HOURS BEFORE BED CAN REDUCE THE IMPACT. EXERCISE REGULAR EXERCISE IMPROVES SLEEP QUALITY, INCREASES TIME SPENT IN DEEP SLEEP, AND REDUCES THE TIME IT TAKES TO FALL ASLEEP. THE EFFECTS ARE DOSE-DEPENDENT — MORE EXERCISE IS GENERALLY BETTER, BUT EVEN MODERATE EXERCISE LIKE WALKING HAS BENEFITS. TIMING MATTERS, BUT LESS THAN MOST PEOPLE THINK. MODERATE EXERCISE AT ANY TIME OF DAY IMPROVES SLEEP. THE OLD ADVICE TO AVOID EXERCISE WITHIN THREE HOURS OF BED IS BASED ON THE IDEA THAT EXERCISE IS STIMULATING. FOR SOME PEOPLE, PARTICULARLY THOSE WITH ANXIETY, VIGOROUS EVENING EXERCISE CAN AFFECT SLEEP. FOR OTHERS, IT HAS NO NEGATIVE EFFECT. PAY ATTENTION TO YOUR OWN RESPONSE. THE BEDROOM ENVIRONMENT YOUR BEDROOM SHOULD BE OPTIMIZED FOR SLEEP. IT SHOULD BE DARK, COOL, AND QUIET. YOUR MATTRESS, PILLOWS, AND BEDDING SHOULD BE COMFORTABLE. YOUR BED SHOULD BE RESERVED FOR SLEEP AND INTIMACY ONLY — NOT FOR WORK, EATING, OR WATCHING TV. THE

ASSOCIATION BETWEEN YOUR BED AND SLEEP IS IMPORTANT. IF YOU SPEND HOURS IN BED WATCHING NETFLIX OR SCROLLING THROUGH YOUR PHONE, YOUR BRAIN LEARNS THAT BED IS A PLACE FOR WAKEFUL ACTIVITIES. STIMULUS CONTROL THERAPY, MENTIONED IN CHAPTER 3, IS BUILT ON THE PRINCIPLE OF STRENGTHENING THE BED-SLEEP ASSOCIATION. SLEEP HYGIENE IS NOT A CURE ONE FINAL NOTE: SLEEP HYGIENE IS IMPORTANT, BUT IT IS NOT A CURE FOR SLEEP DISORDERS. PEOPLE WITH CHRONIC INSOMNIA OFTEN HAVE EXCELLENT SLEEP HYGIENE. THE PROBLEM IS NOT THEIR HABITS — IT IS THEIR NERVOUS SYSTEM. SLEEP HYGIENE IS A FOUNDATION, NOT A TREATMENT. IF YOU HAVE A DIAGNOSED SLEEP DISORDER, SLEEP HYGIENE ALONE WILL NOT RESOLVE IT. BUT IMPROVING YOUR SLEEP HYGIENE WILL MAKE OTHER TREATMENTS MORE EFFECTIVE. REFLECTION QUESTIONS

Which of these sleep hygiene factors is your strongest area? Which needs the most improvement? How consistent is your sleep schedule, including weekends? What is one change to your sleep environment that you could make this week? Have you noticed a relationship between your caffeine or alcohol intake and your sleep quality?

Napping: Helpful or Harmful? Napping is a complex topic in sleep medicine. For some people, a strategic nap can improve alertness and performance. For others, napping fragments nighttime sleep and worsens insomnia. The key factors are timing and duration. The ideal nap is short, typically ten to twenty minutes, and occurs in the early afternoon. This length is enough to improve alertness without entering deep sleep, which w

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CHAPTER 10: CBT-I AND OTHER EVIDENCE-BASED TREATMENTS

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WHEN SLEEP PROBLEMS BECOME CHRONIC, BASIC SLEEP HYGIENE IS RARELY ENOUGH. THE NEXT LEVEL OF TREATMENT INVOLVES STRUCTURED, EVIDENCE-BASED APPROACHES THAT TARGET THE UNDERLYING MECHANISMS OF SLEEP DISORDERS. THE MOST EFFECTIVE AND WELL-STUDIED OF THESE IS COGNITIVE BEHAVIORAL THERAPY FOR INSOMNIA, OR CBT-I. BUT THERE ARE OTHER APPROACHES WORTH UNDERSTANDING AS WELL. WHAT IS CBT-I? COGNITIVE BEHAVIORAL THERAPY FOR INSOMNIA IS A STRUCTURED, MULTI-COMPONENT TREATMENT THAT ADDRESSES THE THOUGHTS, BEHAVIORS, AND EMOTIONS THAT PERPETUATE INSOMNIA. IT IS NOT A SINGLE TECHNIQUE BUT A COMBINATION OF SEVERAL INTERVENTIONS, EACH TARGETING A DIFFERENT ASPECT OF THE INSOMNIA CYCLE. CBT-I IS TYPICALLY DELIVERED OVER FOUR TO EIGHT SESSIONS WITH A TRAINED THERAPIST, BUT THERE ARE ALSO EFFECTIVE SELF-GUIDED PROGRAMS AND DIGITAL VERSIONS. THE CORE COMPONENTS INCLUDE SLEEP RESTRICTION THERAPY, STIMULUS CONTROL THERAPY, COGNITIVE THERAPY, AND RELAXATION TRAINING. RESEARCH HAS CONSISTENTLY FOUND THAT CBT-I IS EFFECTIVE FOR 70 TO 80 PERCENT OF PEOPLE WITH CHRONIC INSOMNIA. IT PRODUCES IMPROVEMENTS THAT ARE COMPARABLE TO SLEEP MEDICATIONS IN THE SHORT TERM AND SUPERIOR IN THE LONG TERM. MOST IMPORTANTLY, THE BENEFITS PERSIST AFTER TREATMENT ENDS, UNLIKE MEDICATIONS THAT STOP WORKING WHEN YOU STOP TAKING THEM. SLEEP RESTRICTION

THERAPY SLEEP RESTRICTION THERAPY IS THE MOST POWERFUL BEHAVIORAL COMPONENT OF CBT-I. IT SOUNDS COUNTERINTUITIVE: TO SLEEP BETTER, YOU SPEND LESS TIME IN BED. THE LOGIC IS THAT PEOPLE WITH INSOMNIA OFTEN SPEND EXCESSIVE TIME IN BED TRYING TO CATCH UP ON SLEEP. THIS BEHAVIOR FRAGMENTS SLEEP, REDUCES SLEEP DRIVE, AND WEAKENS THE ASSOCIATION BETWEEN BED AND SLEEP. IN SLEEP RESTRICTION THERAPY, YOU CALCULATE YOUR AVERAGE TOTAL SLEEP TIME FROM A SLEEP DIARY, THEN LIMIT YOUR TIME IN BED TO THAT AMOUNT. FOR EXAMPLE, IF YOU AVERAGE FIVE HOURS OF SLEEP BUT SPEND EIGHT HOURS IN BED, YOU WOULD INITIALLY RESTRICT YOUR TIME IN BED TO FIVE AND A HALF HOURS. THIS CREATES A MILD SLEEP DEBT THAT INCREASES YOUR SLEEP DRIVE. YOU FALL ASLEEP FASTER, WAKE UP LESS DURING THE NIGHT, AND SLEEP MORE DEEPLY. AS YOUR SLEEP EFFICIENCY IMPROVES — MEANING THE PERCENTAGE OF TIME IN BED THAT YOU ACTUALLY SLEEP — YOUR TIME IN BED IS GRADUALLY INCREASED. SLEEP RESTRICTION THERAPY IS EFFECTIVE, BUT IT IS CHALLENGING. THE FIRST FEW WEEKS ARE DIFFICULT BECAUSE YOU ARE DELIBERATELY RESTRICTING YOUR SLEEP. IT IS IMPORTANT TO DO THIS UNDER PROFESSIONAL GUIDANCE, PARTICULARLY IF YOU OPERATE HEAVY MACHINERY OR DRIVE FOR WORK.

STIMULUS CONTROL THERAPY STIMULUS CONTROL THERAPY IS BASED ON THE IDEA THAT INSOMNIA INVOLVES A LEARNED ASSOCIATION BETWEEN THE BEDROOM AND WAKEFULNESS. THE BED BECOMES A PLACE OF FRUSTRATION, WORRY, AND ALERTNESS RATHER THAN RELAXATION AND SLEEP. STIMULUS CONTROL THERAPY AIMS TO REVERSE THIS ASSOCIATION. THE INSTRUCTIONS ARE SIMPLE BUT CHALLENGING. GO TO BED ONLY WHEN YOU ARE SLEEPY. IF YOU CANNOT FALL ASLEEP WITHIN TWENTY MINUTES, GET OUT OF BED. GO TO ANOTHER ROOM AND DO SOMETHING QUIET AND RELAXING UNTIL

YOU FEEL SLEEPY, THEN RETURN TO BED. IF YOU STILL CANNOT SLEEP, GET UP AGAIN. REPEAT AS NEEDED. THIS TECHNIQUE IS DIFFICULT BECAUSE IT REQUIRES GETTING OUT OF BED WHEN ALL YOU WANT TO DO IS STAY IN BED AND TRY HARDER TO SLEEP. BUT IT WORKS. OVER TIME, YOUR BRAIN RELEARNS THAT BED IS A PLACE FOR SLEEP, AND THE CONDITIONED AROUSAL FADES. COGNITIVE THERAPY COGNITIVE THERAPY IN CBT-I TARGETS THE THOUGHTS AND BELIEFS THAT FUEL INSOMNIA. COMMON UNHELPFUL BELIEFS INCLUDE CATASTROPHIC PREDICTIONS ABOUT THE CONSEQUENCES OF POOR SLEEP, UNREALISTIC EXPECTATIONS ABOUT HOW MUCH SLEEP YOU NEED, AND A SENSE OF HELPLESSNESS ABOUT YOUR ABILITY TO SLEEP. THE COGNITIVE THERAPIST HELPS YOU IDENTIFY THESE THOUGHTS AND CHALLENGE THEM WITH EVIDENCE. FOR EXAMPLE, IF YOU BELIEVE THAT YOU CANNOT FUNCTION AFTER A BAD NIGHT OF SLEEP, YOU MIGHT EXAMINE THE EVIDENCE. HAS THERE EVER BEEN A DAY WHEN YOU FUNCTIONED OKAY DESPITE POOR SLEEP? WHAT IS THE ACTUAL IMPACT VERSUS THE PREDICTED IMPACT? THIS COGNITIVE RESTRUCTURING REDUCES THE ANXIETY ABOUT SLEEP, WHICH IN TURN REDUCES THE HYPERAROUSAL THAT PERPETUATES INSOMNIA. RELAXATION TRAINING RELAXATION TRAINING ADDRESSES THE PHYSIOLOGICAL HYPERAROUSAL THAT CHARACTERIZES CHRONIC INSOMNIA. TECHNIQUES INCLUDE PROGRESSIVE MUSCLE RELAXATION, DIAPHRAGMATIC BREATHING, GUIDED IMAGERY, AND MINDFULNESS MEDITATION. PROGRESSIVE MUSCLE RELAXATION INVOLVES TENSING AND THEN RELEASING DIFFERENT MUSCLE GROUPS IN SEQUENCE. THIS CREATES A PHYSICAL SENSATION OF RELAXATION THAT COUNTERACTS THE TENSION OF HYPERAROUSAL. DIAPHRAGMATIC BREATHING ACTIVATES THE PARASYMPATHETIC NERVOUS SYSTEM, LOWERING HEART RATE AND BLOOD PRESSURE. MINDFULNESS MEDITATION

TEACHES YOU TO OBSERVE THOUGHTS WITHOUT ENGAGING WITH THEM, REDUCING THE IMPACT OF RACING THOUGHTS AT BEDTIME. MINDFULNESS-BASED APPROACHES MINDFULNESS-BASED TREATMENTS FOR INSOMNIA, OR MBTI, COMBINE MINDFULNESS MEDITATION WITH SLEEP-SPECIFIC CONTENT. THE APPROACH TEACHES YOU TO RELATE DIFFERENTLY TO THE EXPERIENCE OF BEING AWAKE AT NIGHT. INSTEAD OF FIGHTING WAKEFULNESS, YOU OBSERVE IT WITH ACCEPTANCE AND NON-JUDGMENT. RESEARCH HAS FOUND THAT MBTI IS EFFECTIVE FOR INSOMNIA, PARTICULARLY FOR THE COGNITIVE ASPECTS. IT REDUCES THE RUMINATION AND WORRY THAT KEEP PEOPLE AWAKE, AND IT DECREASES THE EMOTIONAL DISTRESS ASSOCIATED WITH POOR SLEEP. LIGHT THERAPY AS DISCUSSED IN CHAPTER 6, LIGHT THERAPY IS THE PRIMARY TREATMENT FOR CIRCADIAN RHYTHM DISORDERS. BRIGHT LIGHT EXPOSURE IN THE MORNING ADVANCES THE CIRCADIAN CLOCK, MAKING IT EASIER TO FALL ASLEEP EARLIER. LIGHT BOXES THAT DELIVER 10,000 LUX ARE THE STANDARD TOOL. TREATMENT TYPICALLY INVOLVES THIRTY TO SIXTY MINUTES OF EXPOSURE SHORTLY AFTER WAKING. LIGHT THERAPY CAN ALSO BE USED IN COMBINATION WITH CBT-I FOR PEOPLE WHO HAVE BOTH INSOMNIA AND A DELAYED CIRCADIAN RHYTHM. BIOFEEDBACK BIOFEEDBACK IS A TECHNIQUE THAT TEACHES YOU TO CONTROL PHYSIOLOGICAL PROCESSES THAT ARE NORMALLY AUTOMATIC. SENSORS MEASURE HEART RATE, MUSCLE TENSION, SKIN TEMPERATURE, OR BRAIN WAVES, AND PROVIDE REAL-TIME FEEDBACK. YOU LEARN TO CONSCIOUSLY CHANGE THESE RESPONSES THROUGH RELAXATION AND MENTAL TECHNIQUES. BIOFEEDBACK HAS BEEN SHOWN TO BE EFFECTIVE FOR INSOMNIA, PARTICULARLY WHEN COMBINED WITH OTHER BEHAVIORAL INTERVENTIONS. IT IS LESS WIDELY AVAILABLE THAN CBT-I BUT CAN BE A USEFUL OPTION FOR PEOPLE WHO DO NOT

RESPOND TO OTHER TREATMENTS. CHOOSING THE RIGHT TREATMENT THE CHOICE OF TREATMENT DEPENDS ON YOUR SPECIFIC SLEEP DISORDER, ITS SEVERITY, YOUR PREFERENCES, AND YOUR ACCESS TO HEALTHCARE. FOR CHRONIC INSOMNIA, CBT-I IS THE FIRST-LINE TREATMENT RECOMMENDED BY EVERY MAJOR SLEEP MEDICINE ORGANIZATION. FOR CIRCADIAN RHYTHM DISORDERS, LIGHT THERAPY IS THE PRIMARY APPROACH. FOR SLEEP APNEA, CPAP IS THE GOLD STANDARD. FOR RESTLESS LEGS SYNDROME, LIFESTYLE CHANGES AND MEDICATION ARE THE MAIN OPTIONS. WHATEVER TREATMENT YOU PURSUE, THE MOST IMPORTANT FACTOR IS CONSISTENCY. SLEEP TREATMENTS TAKE TIME AND REQUIRE COMMITMENT. IMPROVEMENT IS RARELY LINEAR, BUT FOR MOST PEOPLE, IT IS INEVITABLE IF YOU STICK WITH THE TREATMENT.

REFLECTION QUESTIONS

Which component of CBT-I do you think would be most challenging for you? Which sounds most helpful? Have you ever tried any structured treatment for your sleep problems? What was your experience? What would need to change in your life for you to prioritize a sleep treatment program? If you could only make one change to improve your sleep, what would it be?

Combining Treatments For many people with sleep disorders, a combination of treatments works better than any single approach. Combining CBT-I with light therapy can address both the behavioral and circadian aspects of insomnia. Combining CPAP with weight loss and positional therapy can address multiple factors in sleep apnea. The key is to start with one evidence-based treatment and add additional intervent

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CHAPTER 11: BUILDING YOUR PERSONAL SLEEP RECOVERY PLAN

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YOU HAVE NOW LEARNED ABOUT THE SCIENCE OF SLEEP, THE MAJOR SLEEP DISORDERS, AND THE TREATMENTS THAT CAN HELP. KNOWLEDGE IS VALUABLE, BUT IT IS NOT ENOUGH. THE FINAL STEP IS TRANSLATING WHAT YOU HAVE LEARNED INTO A PERSONAL ACTION PLAN THAT FITS YOUR LIFE, YOUR SLEEP PROBLEMS, AND YOUR GOALS. THIS CHAPTER WILL GUIDE YOU THROUGH BUILDING THAT PLAN. IT IS NOT A ONE-SIZE-FITS-ALL PRESCRIPTION. IT IS A FRAMEWORK FOR CREATING YOUR OWN PATH TO BETTER SLEEP.

STEP ONE: UNDERSTAND YOUR SLEEP PROBLEM

THE FIRST STEP IS CLARITY. WHAT EXACTLY IS GOING WRONG WITH YOUR SLEEP? BE SPECIFIC. DO YOU HAVE TROUBLE FALLING ASLEEP? DO YOU WAKE UP IN THE MIDDLE OF THE NIGHT? DO YOU WAKE UP FEELING UNREFRESHED EVEN AFTER WHAT SEEMS LIKE ADEQUATE SLEEP? DO YOU SNORE? DO YOU HAVE RESTLESS LEGS? DO YOU FEEL LIKE YOUR BODY CLOCK IS OUT OF SYNC? IF YOU ARE NOT SURE, KEEP A SLEEP DIARY FOR TWO WEEKS. RECORD WHEN YOU GO TO BED, WHEN YOU FALL ASLEEP, HOW OFTEN YOU WAKE UP, WHEN YOU WAKE UP FOR THE DAY, AND HOW RESTED YOU FEEL. NOTE ANY SYMPTOMS LIKE SNORING, GASPING, LEG MOVEMENTS, OR UNUSUAL BEHAVIORS. THIS DIARY WILL REVEAL PATTERNS THAT YOU MIGHT NOT NOTICE IN THE MOMENT. IT WILL ALSO PROVIDE VALUABLE INFORMATION IF YOU DECIDE TO CONSULT A SLEEP SPECIALIST.

STEP TWO: IDENTIFY YOUR PRIORITIES BASED ON YOUR UNDERSTANDING OF YOUR SLEEP PROBLEM, IDENTIFY YOUR TOP PRIORITIES. IF YOU

SUSPECT SLEEP APNEA, YOUR PRIORITY MIGHT BE GETTING A SLEEP STUDY. IF YOU HAVE CHRONIC INSOMNIA, YOUR PRIORITY MIGHT BE FINDING A CBT-I PROVIDER. IF YOU HAVE A CIRCADIAN RHYTHM DISORDER, YOUR PRIORITY MIGHT BE IMPLEMENTING MORNING LIGHT THERAPY. YOUR PRIORITIES SHOULD BE REALISTIC AND SPECIFIC. INSTEAD OF "I WANT TO SLEEP BETTER," SET A GOAL LIKE "I WANT TO REDUCE THE TIME IT TAKES ME TO FALL ASLEEP FROM NINETY MINUTES TO THIRTY MINUTES WITHIN EIGHT WEEKS." STEP THREE: START WITH THE FOUNDATION BEFORE IMPLEMENTING ANY SPECIFIC TREATMENT, MAKE SURE THE FOUNDATION IS SOLID. THIS MEANS ADDRESSING BASIC SLEEP HYGIENE — CONSISTENCY, LIGHT EXPOSURE, TEMPERATURE, NOISE, CAFFEINE, ALCOHOL, AND EXERCISE. THESE ARE NOT TREATMENTS FOR SLEEP DISORDERS, BUT THEY ARE THE FOUNDATION ON WHICH TREATMENTS ARE BUILT. IF YOUR FOUNDATION IS WEAK, EVEN THE BEST TREATMENTS WILL BE LESS EFFECTIVE. TAKE A WEEK TO OPTIMIZE YOUR SLEEP HYGIENE BEFORE MOVING ON TO MORE ADVANCED INTERVENTIONS. STEP FOUR: CHOOSE YOUR INTERVENTIONS BASED ON YOUR SPECIFIC SLEEP PROBLEM, CHOOSE ONE OR TWO INTERVENTIONS TO START WITH. DO NOT TRY TO IMPLEMENT EVERYTHING AT ONCE. OVERWHELMING YOURSELF WILL LEAD TO INCONSISTENCY, AND INCONSISTENCY LEADS TO FAILURE. IF YOU HAVE INSOMNIA, START WITH STIMULUS CONTROL OR SLEEP RESTRICTION. IF YOU HAVE A CIRCADIAN RHYTHM DISORDER, START WITH MORNING LIGHT THERAPY. IF YOU HAVE RESTLESS LEGS SYNDROME, START WITH THE LIFESTYLE MODIFICATIONS AND CHECK YOUR IRON LEVELS. IF YOU HAVE SLEEP APNEA, START WITH THE DIAGNOSTIC PROCESS AND CPAP THERAPY. GIVE EACH INTERVENTION AT LEAST TWO TO FOUR WEEKS BEFORE EVALUATING ITS EFFECTIVENESS. SLEEP

TREATMENTS TAKE TIME. DO NOT GIVE UP AFTER A FEW NIGHTS. STEP FIVE: TRACK YOUR PROGRESS CONTINUE KEEPING A SLEEP DIARY AS YOU IMPLEMENT YOUR INTERVENTIONS. THIS WILL SHOW YOU WHAT IS WORKING AND WHAT IS NOT. LOOK FOR TRENDS OVER WEEKS, NOT NIGHTS. A SINGLE BAD NIGHT DOES NOT MEAN THE TREATMENT IS FAILING. A PATTERN OF IMPROVEMENT OVER SEVERAL WEEKS IS MEANINGFUL. ALSO TRACK YOUR DAYTIME FUNCTION — YOUR ENERGY LEVELS, MOOD, CONCENTRATION, AND OVERALL WELL-BEING. THE GOAL OF SLEEP TREATMENT IS NOT JUST BETTER SLEEP BUT A BETTER LIFE. IF YOUR SLEEP IMPROVES BUT YOUR DAYTIME FUNCTION DOES NOT, THERE MAY BE OTHER FACTORS AT PLAY. STEP SIX: ADJUST AND ITERATE TREATMENT IS NOT A LINEAR PROCESS. IT INVOLVES EXPERIMENTATION AND ADJUSTMENT. IF A PARTICULAR INTERVENTION IS NOT WORKING, TRY A DIFFERENT ONE. IF IT IS WORKING BUT SLOWLY, GIVE IT MORE TIME. IF YOU HIT A PLATEAU, CONSIDER ADDING A SECOND INTERVENTION. BE PATIENT WITH YOURSELF. SLEEP DISORDERS DEVELOP OVER MONTHS OR YEARS, AND THEY DO NOT RESOLVE OVERNIGHT. BUT WITH CONSISTENT EFFORT, THEY DO RESOLVE. STEP SEVEN: KNOW WHEN TO GET PROFESSIONAL HELP SOME SLEEP PROBLEMS REQUIRE PROFESSIONAL INTERVENTION. IF YOU SUSPECT SLEEP APNEA, IF YOUR INSOMNIA IS SEVERE OR LONG-STANDING, IF YOU HAVE SYMPTOMS OF RESTLESS LEGS SYNDROME THAT ARE SIGNIFICANTLY AFFECTING YOUR QUALITY OF LIFE, OR IF YOU EXPERIENCE DANGEROUS PARASOMNIAS, CONSULT A HEALTHCARE PROVIDER. A SLEEP SPECIALIST CAN PROVIDE DIAGNOSTIC TESTING, PRESCRIBE TREATMENTS THAT ARE NOT AVAILABLE OVER THE COUNTER, AND GUIDE YOU THROUGH THE PROCESS WITH EXPERTISE THAT SELF-HELP CANNOT MATCH. THERE IS NO SHAME IN SEEKING

HELP. SLEEP DISORDERS ARE MEDICAL CONDITIONS, AND THEY DESERVE MEDICAL ATTENTION. THE BIG PICTURE IMPROVING YOUR SLEEP IS ONE OF THE MOST IMPORTANT INVESTMENTS YOU CAN MAKE IN YOUR HEALTH. BETTER SLEEP IMPROVES YOUR MOOD, SHARPENS YOUR THINKING, STRENGTHENS YOUR IMMUNE SYSTEM, REDUCES YOUR RISK OF CHRONIC DISEASE, AND ENHANCES YOUR QUALITY OF LIFE. BUT IT IS ALSO IMPORTANT TO KEEP PERSPECTIVE. PERFECT SLEEP IS NOT THE GOAL. EVERYONE HAS BAD NIGHTS. THE GOAL IS TO MOVE FROM CHRONIC, DEBILITATING SLEEP PROBLEMS TO A PLACE WHERE SLEEP IS GENERALLY GOOD AND OCCASIONAL BAD NIGHTS ARE MANAGEABLE. THIS IS ACHIEVABLE. THE SCIENCE IS CLEAR. THE TREATMENTS EXIST. THE PATH IS KNOWN. WHAT REMAINS IS YOUR COMMITMENT TO WALKING IT. A FINAL REFLECTION YOU PICKED UP THIS BOOK BECAUSE SOMETHING ABOUT YOUR SLEEP WAS NOT WORKING. MAYBE YOU WERE EXHAUSTED. MAYBE YOU WERE FRUSTRATED. MAYBE YOU WERE WORRIED. WHATEVER BROUGHT YOU HERE, I WANT TO ACKNOWLEDGE THE COURAGE IT TAKES TO CONFRONT A PROBLEM THAT AFFECTS YOU SO INTIMATELY AND SO PERSISTENTLY. SLEEP IS PERSONAL. IT IS VULNERABLE. STRUGGLING WITH SLEEP CAN FEEL LIKE A FAILURE OF SOMETHING FUNDAMENTAL. IT IS NOT. IT IS A MEDICAL CONDITION, LIKE ANY OTHER, AND IT DESERVES THE SAME COMPASSION AND CARE. THE ROAD TO BETTER SLEEP IS NOT ALWAYS STRAIGHT, BUT IT IS ALWAYS WORTH TRAVELING. EVERY STEP YOU TAKE — EVERY HABIT YOU ADJUST, EVERY TREATMENT YOU TRY, EVERY NIGHT YOU COMMIT TO CONSISTENCY — BRINGS YOU CLOSER TO THE REST YOU DESERVE. YOU DESERVE TO SLEEP WELL. YOU DESERVE TO WAKE UP FEELING RESTORED. YOU DESERVE THE PEACE THAT COMES FROM KNOWING YOUR BODY AND MIND ARE FUNCTIONING AS THEY SHOULD. NOW GO. START YOUR JOURNEY.

ONE NIGHT AT A TIME. PRACTICAL EXERCISE: YOUR SLEEP RECOVERY PLAN TAKE A PIECE OF PAPER OR OPEN A DOCUMENT ON YOUR PHONE. WRITE DOWN THE FOLLOWING:

My main sleep problem in one sentence: _ My sleep goal for the next eight weeks: _ One sleep hygiene change I will implement this week: _ One treatment intervention I will start: _ When I will review my progress: _ Who I will talk to if I need professional help: _

Keep this plan somewhere visible. Refer to it regularly. And celebrate every small victory along the way. Dealing with Setbacks Setbacks are a normal part of any behavior change journey. You will have bad nights. You will have weeks where your sleep worsens for no apparent reason. You will have times when you fall back into old habits. None of this means you have failed. The key is how you respond to setbacks. If you have a bad night, do not catastrophize. A single bad night does not undo weeks of progress. If you fall back into an old habit, notice it without judgment and return to your plan the next day. The most common mistake people make after a setback is trying harder. They increase their effort, become more rigid, and create more anxiety around sleep. The better response is to return to the basics — consistent schedule, good sleep hygiene, and the behavioral techniques you have learned. Building Long-Term Maintenance The goal of treatment is not just to improve your sleep temporarily but to maintain those improvements over the long term. Building maintenance into your plan

resources

RESOURCES AND RECOMMENDED READING
RESOURCES AND RECOMMENDED READING THIS SECTION PROVIDES A CURATED LIST OF RESOURCES FOR FURTHER EXPLORATION. THE BOOKS, ORGANIZATIONS, AND WEBSITES LISTED HERE ARE WELL-REGARDED IN THE FIELD OF SLEEP MEDICINE AND HAVE BEEN VETTED FOR ACCURACY AND USEFULNESS. BOOKS

Why We Sleep by Matthew Walker, PhD — A comprehensive and accessible overview of sleep science. Walker explains the functions of sleep, the consequences of sleep deprivation, and practical strategies for improving sleep. This is the most widely recommended book on sleep for general readers. The Sleep Solution by W. Chris Winter, MD — A practical, evidence-based guide to understanding and treating sleep problems. Winter offers clear explanations of sleep disorders and actionable advice for improving sleep. Quiet Your Mind and Get to Sleep by Colleen Carney, PhD, and Rachel Manber, PhD — A CBT-I workbook that provides structured exercises for managing insomnia. This is an excellent resource for anyone who wants to implement CBT-I techniques on their own. The Circadian Code by Satchin Panda, PhD — An exploration of circadian rhythms and how they affect health. Panda explains how light, food, and activity timing influence the body clock and offers practical advice for alignment. Say Good Night to Insomnia by Gregg

D. Jacobs, PhD — A comprehensive guide to CBT-I that has been validated in clinical trials. The book provides a step-by-step program for overcoming chronic insomnia.

Organizations

American Academy of Sleep Medicine — The professional organization for sleep medicine specialists. Their website provides information on sleep disorders, treatment guidelines, and a directory of accredited sleep centers. National Sleep Foundation — An independent nonprofit that provides education and advocacy for sleep health. Their website offers evidence-based information on sleep disorders, sleep hygiene, and the latest research. Sleep Research Society — A scientific organization that advances sleep research. Their publications and conferences drive the field forward. Restless Legs Syndrome Foundation — A nonprofit dedicated to RLS research, education, and advocacy. Their website provides resources for patients and providers.

Websites and Online Resources

SleepEducation.org — The patient education website of the American Academy of Sleep Medicine. It provides clear, reliable information on sleep disorders and treatments. CBT-I Coach — A free mobile app developed by the Department of Veterans Affairs that guides users through the core components of CBT-I. The app includes sleep tracking, relaxation exercises, and educational content. Insomnia Coach — Another free app from the VA that provides CBT-I tools and tracking for people with insomnia. MyCBT — A self-

guided online CBT-I program that has been validated in clinical research. It provides structured lessons, sleep tracking, and personalized feedback.

For Healthcare Providers

International Classification of Sleep Disorders, Third Edition — The standard diagnostic manual for sleep disorders. It provides detailed diagnostic criteria for all recognized sleep disorders. **Cognitive Behavioral Therapy for Insomnia: A Comprehensive Guide for Clinicians** — A clinical resource for healthcare providers who want to deliver CBT-I in their practice.

When to Seek Professional Help The following situations warrant consultation with a healthcare provider or sleep specialist:

Loud, disruptive snoring, particularly if accompanied by pauses in breathing
Excessive daytime sleepiness that interferes with daily activities
Falling asleep involuntarily in situations that require alertness
Uncomfortable sensations in the legs that interfere with sleep
Acting out dreams or engaging in complex behaviors during sleep
Insomnia that persists for more than three months despite self-help efforts
Sleep problems that significantly affect your quality of life

A Note on Statistics The statistics cited in this book are drawn from published research and public health data. Where specific studies are referenced, they have been selected for their methodological rigor and relevance. For the most current statistics, consult the websites of the organizations listed above. **Final Thoughts** Sleep is not a luxury. It is a biological necessity. The research is clear, the

treatments are effective, and the path to better sleep is known. What remains is the commitment to walk that path. Whether you are dealing with insomnia, sleep apnea, restless legs, circadian rhythm disruption, or any other sleep disorder, there is help available. You do not have to suffer in silence. You do not have to accept poor sleep as your lot in life. The journey starts with a single night. Make it count.

Recommended Sleep Tracking Tools

Keeping a sleep diary is the simplest and most reliable way to track your sleep. A paper diary can be as simple as noting your bedtime, wake time, estimated sleep duration, and sleep quality each morning. Several validated sleep diaries are available online, including the Consensus Sleep Diary. Wearable sleep trackers, such as those from Fitbit, Garmin, and Apple, can provide additional data about sleep duration, sleep stages, and heart rate variability. These devices are not as accurate as laboratory polysomnography, but they can reveal patterns over time. The most useful metrics from wearables are sleep duration and consistency — the stage detection algorithms are less reliable. Smartphone apps that use the phone's microphone and accelerometer to estimate sleep are less accurate than wearables but can still be useful for tracking patterns. Some apps also provide alarm clocks that wake you during light sleep, which can make waking up feel less jarring.

The Importance of a Sleep-Friendly Lifestyle

Improving sleep often requires looking beyond the immediate sleep environment and considering your lifestyle as a whole. Stress management, social connection, physical activity, and meaningful engagement all affect sleep quality indirectly. Chronic stress is one of the most powerful disruptors of sleep. Finding effective ways to manage stress — whether

through therapy, exercise, meditation, hobbies, or social support — can have a profound impact on sleep. The goal is not to eliminate stress, which is impossible, but to build resilience and recovery into your daily life. Physical activity improves sleep quality directly and indirectly. Directly, exercise increases sleep drive and promotes deep sleep. Indirectly, exercise reduces stress, improves mood, and helps regulate energy levels. Even moderate activity like walking has measurable benefits.

A Final Note on Persistence If you have tried multiple approaches to improve your sleep and none have worked, you may feel discouraged. I want to acknowledge that frustration and validate it. Sleep disorders can be stubborn, and finding the right treatment can take time. But do not give up. Sleep science is advancing rapidly. New treatments are being developed. The understanding of sleep is deeper today than it was ten years ago, and it will be deeper still in ten more years. If today's treatments do not work for you, tomorrow's might. In the meantime, continue building the foundation. Consistency, good sleep hygiene, stress management, and physical activity are never wasted effort. They improve your health regardless of whether they resolve your sleep disorder on their own. Your sleep matters. Your health matters. Your quality of life matters. Do not stop fighting for them. Sleep well.

About the Author

AUTHOR BIO GOES HERE

Also by AUTHOR_NAME

This is her first book.

